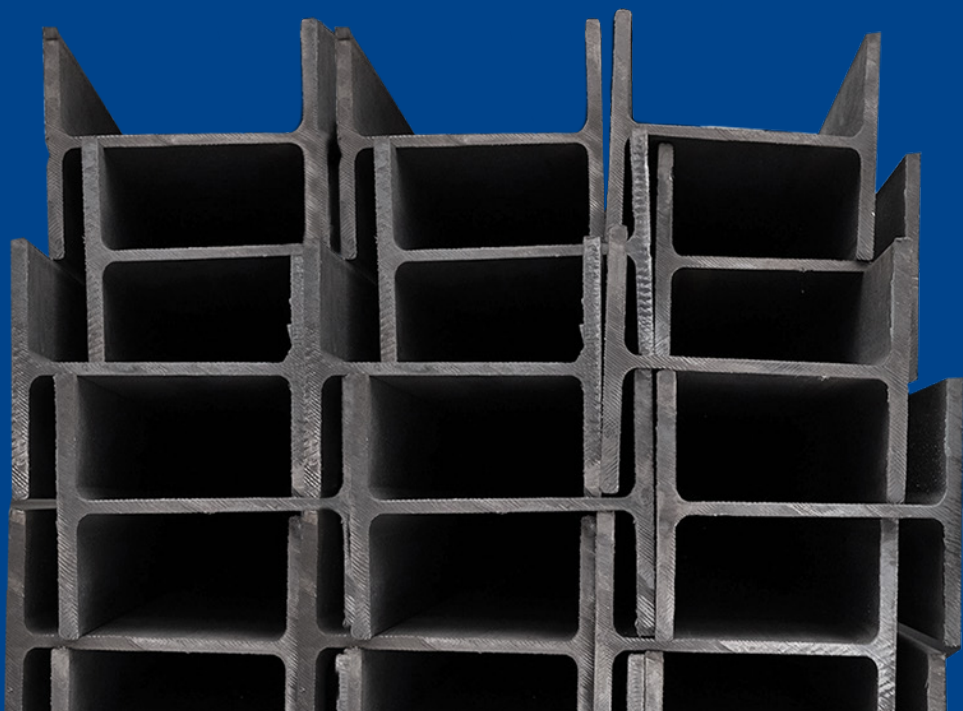




# Catálogo de **Productos**

**Estructurales • Comerciales • Planos • Galvanizados  
Corrugados y Trefilados • Perfiles Tubulares y Tuberías**





**45 años**  
siendo parte de las historias  
que construyen a México



## ÍNDICE

|                                           |           |
|-------------------------------------------|-----------|
| <b>Alambres</b>                           | <b>03</b> |
| <b>Ángulos</b>                            | <b>04</b> |
| <b>Canales</b>                            | <b>05</b> |
| <b>Castillos</b>                          | <b>06</b> |
| <b>Clavos</b>                             | <b>07</b> |
| <b>Cuadrados</b>                          | <b>08</b> |
| <b>HSS Cuadrado</b>                       | <b>09</b> |
| <b>HSS Rectangular</b>                    | <b>10</b> |
| <b>Lámina Galvanizada</b>                 | <b>11</b> |
| <b>Lámina Negra</b>                       | <b>12</b> |
| <b>Malla</b>                              | <b>13</b> |
| <b>Polín Mon-Ten</b>                      | <b>15</b> |
| <b>Placa de Acero</b>                     | <b>17</b> |
| <b>PTR</b>                                | <b>18</b> |
| <b>Redondo</b>                            | <b>08</b> |
| <b>Solera</b>                             | <b>20</b> |
| <b>Tee Y Zeta</b>                         | <b>08</b> |
| <b>Tubería Cédula 30-40</b>               | <b>21</b> |
| <b>Tubería Para Conducción De Fluidos</b> | <b>22</b> |
| <b>Tubo Estructural</b>                   | <b>23</b> |
| <b>Tubular Pintado</b>                    | <b>24</b> |
| <b>Varilla Corrugada</b>                  | <b>27</b> |
| <b>Viga IPS</b>                           | <b>28</b> |
| <b>Viga IPR</b>                           | <b>29</b> |
| <b>Conversión De Unidades</b>             | <b>32</b> |
| <b>Tabla De Equivalencias</b>             | <b>33</b> |
| <b>Andamios</b>                           | <b>34</b> |
| <b>Puntales</b>                           | <b>35</b> |



## Comprometidos con la solidez de tus proyectos

En Aceros Ocotlán, brindamos soluciones en acero con la calidad, disponibilidad y respaldo que nos distinguen. Nuestra experiencia y cobertura nacional nos permiten ser un aliado confiable para obras de cualquier escala, impulsando el desarrollo con materiales que cumplen los más altos estándares del sector.

## PRODUCTOS



### ESTRUCTURALES

Viga IPR  
Viga IPS  
Ángulos APS  
HSS  
Polín A.R.



### COMERCIALES

Redondos  
Cuadrados  
Ángulos  
Tee Y Zeta  
Pulidos  
Polín Comercial  
Polín Mon-Ten



### PLANOS

Placa A-36  
Placa G-50  
Antiderrapante  
Lámina Negra  
Lámina Galvanizada



### CORRUGADOS Y TREFILADOS

Varilla Corrugada  
Alambrón  
Castillos  
Mallas  
Pulidos



### PERFILES TUBULARES Y TUBERÍAS

C-30/C-40/G-50  
Galvanizada  
De Conducción  
Zintro  
PTR  
Tubular



### GALVANIZADOS

Alambre  
Clavos  
Concertina  
Grapas  
Láminas  
Mallas

LO QUE HACE **FUERTE** A MÉXICO

 [acerosocotlan.mx](http://acerosocotlan.mx)



ALAMBRE RECOCIDO

| No. | Diámetro |        | Peso  |            |
|-----|----------|--------|-------|------------|
|     | mm       | pulg   | kg/m  | rollo (kg) |
| 15  | 1.83     | 0.0720 | 0.021 | 50.00      |
| 16  | 1.59     | 0.0625 | 0.016 | 50.00      |
| 17  | 1.21     | 0.0475 | 0.008 | 50.00      |

ALAMBRÓN

| Diámetro |       | Peso<br>kg/m | Área Nom<br>mm |
|----------|-------|--------------|----------------|
| mm       | pulg  |              |                |
| 5.55     | 0.218 | .0250        | 26.8           |
| 6.35     | 1/4"  | 0.248        | 31.7           |
| 7.94     | 5/16" | 0.388        | 49.5           |
| 9.53     | 3/8"  | 0.559        | 71.3           |

Especificación: NMX-B365 y ASTM A-510

ALAMBRE GALVANIZADO SUAVE

| Calibre | Diámetro (mm) | Metros por rollo |
|---------|---------------|------------------|
| 12.5    | 2.44          | 27               |
| 14.5    | 1.85          | 47               |
| 16      | 1.57          | 65               |
| 18      | 1.22          | 110              |
| 20      | 0.84          | 230              |
| 22      | 0.71          | 321              |

- \* Rollos 1 kg capa de zinc uniforme
- \* Identificador de calibre en cada rollo
- \* Cajas con 20 rollos
- \* Especificación: ASTM A-641

ALAMBRE DE PÚAS

| Denominación | Características                                                                                                             | Peso por rollo (kg) | Largo (m) | Resistencia (kg) | Marca             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------|-----------|------------------|-------------------|
| Vaquero I    | 2 alambres galvanizados trenzados cal 15.5<br>púa de 4 puntas cal 16.5<br>separación entre púas 5"<br>rendimiento 21.3 m/kg | 14.0                | 300       | 454              | De Acero          |
| Corcel III   |                                                                                                                             | 16.7                | 360       | 454              | De Acero          |
|              |                                                                                                                             | 14.8                | 300       |                  |                   |
|              |                                                                                                                             | 17.7                | 360       |                  |                   |
|              |                                                                                                                             | 18.8                | 400       |                  |                   |
|              |                                                                                                                             | 23.5                | 500       |                  |                   |
| Bronco       | 2 alambres galvanizados trenzados cal 12.5<br>púa de 4 puntas cal 14.5<br>separación entre púas 5"<br>rendimiento 10.8 m/kg | 20                  | 216       | 431              | De Acero          |
| Nacional     |                                                                                                                             | 30                  | 324       |                  | Aceros Nacionales |
|              |                                                                                                                             | 34                  | 368       |                  |                   |
|              |                                                                                                                             | 34                  | 368       |                  |                   |
|              |                                                                                                                             | 30                  | 300       |                  |                   |
|              |                                                                                                                             | 34                  | 368       |                  |                   |
| Novillo      |                                                                                                                             | 30                  | 300       |                  | Alambres Potosí   |
| Bravo        |                                                                                                                             |                     |           |                  |                   |

Cumple con la norma ASTM A-121

## ÁNGULOS

| Dimensiones    |                      | Peso  |       |         |         |
|----------------|----------------------|-------|-------|---------|---------|
| pulgadas       | mm                   | kg/m  | largo | kg/pza  | lbs/pie |
| 1/8" x 3/4"    | 3.2 x 19.0           | 0.88  | 6.10  | 5.37    | 0.60    |
| 1/8" x 1"      | 3.2 x 25.4           | 1.19  | 6.10  | 7.26    | 0.80    |
| 1/8" x 1 1/4"  | 3.2 x 31.7           | 1.50  | 6.10  | 9.15    | 1.00    |
| 1/8" x 1 1/2"  | 3.2 x 38.1           | 1.83  | 6.10  | 11.16   | 1.20    |
| 1/8" x 1 3/4"  | 3.2 x 44.5           | 2.14  | 6.10  | 13.05   | 1.44    |
| 1/8" x 2"      | 3.2 x 50.8           | 2.46  | 6.10  | 15.01   | 1.70    |
| 5/32" x 1 1/2" | 4.06 x 38.1          | 2.28  | 6.10  | 13.908  | 1.53    |
| 5/32" x 2"     | 4.06 x 50.8          | 3.09  | 6.10  | 18.849  | 2.07    |
| 5/32" x 2 1/2" | 4.06 x 63.5          | 3.92  | 6.10  | 23.912  | 2.63    |
| 3/16" x 1"     | 4.8 x 25.4           | 1.73  | 6.10  | 10.55   | 1.20    |
| 3/16" x 1 1/4" | 4.8 x 31.7           | 2.20  | 6.10  | 13.42   | 1.50    |
| 3/16" x 1 1/2" | 4.8 x 38.1           | 2.68  | 6.10  | 16.35   | 1.80    |
| 3/16" x 1 3/4" | 4.8 x 44.5           | 3.15  | 6.10  | 19.215  | 2.11    |
| 3/16" x 2"     | 4.8 x 50.8           | 3.63  | 6.10  | 22.14   | 2.40    |
| 3/16" x 2 1/2" | 4.8 x 63.5           | 4.61  | 6.10  | 28.12   | 3.10    |
| 3/16" x 3"     | 4.8 x 76.2           | 5.52  | 12.20 | 67.36   | 3.71    |
| 3/16" x 3 1/2" | 4.8 x 88.9           | 6.55  | 6.10  | 39.955  | 4.40    |
| 1/4" x 1"      | 6.3 x 25.4           | 2.22  | 6.10  | 13.54   | 1.50    |
| 1/4" x 1 1/4"  | 6.3 x 31.7           | 2.86  | 6.10  | 17.45   | 1.90    |
| 1/4" x 1 1/2"  | 6.3 x 38.1           | 3.48  | 6.10  | 21.23   | 2.30    |
| 1/4" x 1 3/4"  | 6.3 x 44.5           | 4.12  | 6.10  | 25.13   | 2.77    |
| 1/4" x 2"      | 6.3 x 50.8           | 4.75  | 6.10  | 28.98   | 3.20    |
| 1/4" x 2 1/2"  | 6.3 x 63.6           | 6.10  | 6.10  | 37.21   | 4.09    |
| 1/4" x 3"      | 6.3 x 76.2           | 7.29  | 12.20 | 88.94   | 4.90    |
| 1/4" x 3 1/2"  | 6.3 x 88.9           | 8.63  | 12.20 | 105.29  | 5.80    |
| 1/4" x 4"      | 6.3 x 101.6          | 9.82  | 12.20 | 119.8   | 6.60    |
| 5/16" x 2"     | 7.9 x 50.8           | 5.83  | 6.10  | 35.56   | 3.91    |
| 5/16" x 2 1/2" | 7.9 x 63.5           | 7.44  | 6.10  | 45.38   | 5.00    |
| 5/16" x 3"     | 7.9 x 76.2           | 9.08  | 12.20 | 110.78  | 6.10    |
| 5/16" x 3 1/2" | 7.9 x 88.9           | 10.71 | 12.20 | 130.66  | 7.20    |
| 5/16" x 4"     | 7.9 x 101.6          | 12.20 | 12.20 | 148.84  | 8.20    |
| 5/16" x 5"     | 7.9 x 127            | 15.47 | 6.10  | 94.367  | 10.39   |
| 3/8" x 2"      | 9.5 x 50.8           | 6.99  | 6.10  | 42.64   | 4.69    |
| 3/8" x 2 1/2"  | 9.5 x 63.5           | 8.78  | 6.10  | 53.56   | 5.90    |
| 3/8" x 3"      | 9.5 x 76.1           | 10.72 | 12.20 | 130.78  | 7.20    |
| 3/8" x 3 1/2"  | 9.5 x 88.9           | 12.64 | 12.20 | 154.33  | 8.50    |
| 3/8" x 4"      | 9.5 x 101.6          | 14.58 | 12.20 | 177.88  | 9.80    |
| 3/8" x 5"      | 9.5 x 127            | 18.30 | 6.10  | 111.63  | 12.28   |
| 3/8" x 6"      | 9.5 x 152.4          | 22.17 | 12.20 | 270.47  | 14.90   |
| 3/8" x 6" x 4" | 9.5 x 152.4 x 101.6  | 18.30 | 12.20 | 223.26  | 12.30   |
| 7/16" x 3 1/2" | 11.11 x 88.9         | 14.58 | 12.20 | 177.88  | 9.80    |
| 7/16" x 4"     | 11.11 x 101.6        | 16.81 | 12.20 | 205.08  | 11.30   |
| 7/16" x 5"     | 11.11 x 127          | 21.28 | 6.10  | 129.808 | 14.29   |
| 7/16" x 6"     | 11.11 x 152.4        | 25.59 | 12.20 | 312.32  | 17.20   |
| 1/2" x 2 1/2"  | 12.7 x 63.5          | 11.83 | 6.10  | 72.16   | 7.95    |
| 1/2" x 3"      | 12.7 x 76.2          | 13.99 | 12.20 | 170.68  | 9.40    |
| 1/2" x 3 1/2"  | 12.7 x 88.9          | 16.51 | 12.20 | 201.42  | 11.10   |
| 1/2" x 4"      | 12.7 x 101.6         | 19.05 | 12.20 | 232.41  | 12.80   |
| 1/2" x 5"      | 12.7 x 127           | 24.11 | 6.10  | 147.071 | 16.19   |
| 1/2" x 6"      | 12.7 x 152.4         | 29.17 | 12.20 | 355.87  | 19.60   |
| 1/2" x 6" x 4" | 12.7 x 152.4 x 101.6 | 24.11 | 12.20 | 294.14  | 16.20   |
| 5/8" x 4"      | 15.9 x 101.6         | 23.38 | 12.20 | 285.24  | 15.71   |
| 5/8" x 6"      | 15.9 x 152.4         | 36.01 | 12.20 | 439.32  | 24.20   |
| 5/8" x 6" x 4" | 15.9 x 152.4 x 101.6 | 29.76 | 12.20 | 363.07  | 20.00   |
| 3/4" x 4"      | 19.0 x 101.6         | 27.53 | 12.20 | 335.87  | 18.50   |
| 3/4" x 6"      | 19.0 x 152.4         | 42.71 | 12.20 | 521.06  | 28.70   |
| 3/4" x 6" x 4" | 19.0 x 152.4 x 101.6 | 35.12 | 12.20 | 428.46  | 23.60   |

| Peralte |       | Peso  |        | Dimensiones |       |            |       |            |       |       |       |
|---------|-------|-------|--------|-------------|-------|------------|-------|------------|-------|-------|-------|
|         |       |       |        | A           |       | B          |       | C          |       |       |       |
|         |       |       |        | Patín       |       | Esp. Patín |       | Esp. Patín |       |       |       |
| pulg    | mm    | kg/m  | kg/pza | pulg        | mm    | pulg       | mm    | pulg       | mm    | pulg  | mm    |
| 3"      | 76.2  | 5.20  | 63.44  | 1.410       | 35.81 | 0.254      | 6.45  | 0.125      | 3.18  | 0.125 | 3.18  |
| 3"      | 76.2  | 6.10  | 74.42  | 1.410       | 35.80 | 0.273      | 6.93  | 0.170      | 4.30  | 0.170 | 4.30  |
| 4"      | 101.6 | 6.69  | 81.62  | 1.580       | 40.23 | 0.220      | 5.60  | 0.174      | 4.42  | 0.174 | 4.42  |
| 4"      | 101.6 | 8.04  | 98.09  | 1.580       | 40.23 | 0.296      | 7.52  | 0.184      | 4.67  | 0.184 | 4.67  |
| 5"      | 127.5 | 9.97  | 121.63 | 1.750       | 44.45 | 0.320      | 8.13  | 0.190      | 4.83  | 0.190 | 4.83  |
| 5"      | 127.5 | 13.39 | 163.36 | 1.885       | 47.88 | 0.320      | 8.13  | 0.325      | 8.26  | 0.325 | 8.26  |
| 6"      | 152.4 | 12.20 | 148.84 | 1.920       | 48.77 | 0.343      | 8.71  | 0.200      | 5.08  | 0.200 | 5.08  |
| 6"      | 152.4 | 15.63 | 190.64 | 2.034       | 51.66 | 0.343      | 8.71  | 0.314      | 7.98  | 0.314 | 7.98  |
| 6"      | 152.4 | 19.35 | 236.03 | 2.157       | 54.79 | 0.343      | 8.71  | 0.437      | 11.10 | 0.437 | 11.10 |
| 8"      | 203.2 | 17.11 | 208.74 | 2.260       | 57.40 | 0.390      | 9.91  | 0.220      | 5.59  | 0.220 | 5.59  |
| 8"      | 203.2 | 20.46 | 249.61 | 2.343       | 59.51 | 0.390      | 9.91  | 0.303      | 7.70  | 0.303 | 7.70  |
| 8"      | 203.2 | 27.90 | 340.38 | 2.527       | 64.19 | 0.390      | 9.91  | 0.487      | 12.37 | 0.487 | 12.37 |
| 10"     | 254.0 | 22.77 | 277.79 | 2.600       | 66.04 | 0.436      | 11.07 | 0.240      | 6.10  | 0.240 | 6.10  |
| 10"     | 254.0 | 29.76 | 363.07 | 2.739       | 69.57 | 0.436      | 11.07 | 0.379      | 9.63  | 0.379 | 9.63  |
| 10"     | 254.0 | 37.20 | 453.84 | 2.886       | 73.30 | 0.436      | 11.07 | 0.526      | 13.36 | 0.526 | 13.36 |
| 10"     | 254.0 | 44.64 | 544.61 | 3.033       | 77.04 | 0.436      | 11.07 | 0.673      | 17.09 | 0.673 | 17.09 |
| 12"     | 304.8 | 30.80 | 375.76 | 2.942       | 74.73 | 0.501      | 12.73 | 0.282      | 7.16  | 0.282 | 7.16  |
| 12"     | 304.8 | 37.20 | 453.84 | 3.047       | 77.39 | 0.501      | 12.73 | 0.387      | 9.83  | 0.387 | 9.83  |
| 12"     | 304.8 | 44.64 | 544.61 | 3.170       | 80.52 | 0.501      | 12.73 | 0.510      | 12.95 | 0.510 | 12.95 |

LONGITUD ESTÁNDAR 12.20 M

CUMPLEN CON LA NORMA ASTM A-36

CASTILLOS ARMEX GRADO 6000

| Características de las Armaduras de Fabricación Estándar                            |                |                        |             |                          |                          |                        |
|-------------------------------------------------------------------------------------|----------------|------------------------|-------------|--------------------------|--------------------------|------------------------|
| Geometría                                                                           | Tipo           | Diámetro de la varilla |             | Sección de concreto (cm) | Sección de armadura (cm) | Peso x pieza de 6.00mt |
|                                                                                     |                | Longitud mm            | Estribos mm |                          |                          |                        |
|  | 12*2           | 6.35                   | 4.115       | 12 X variable            | 8                        | 3.48                   |
|                                                                                     | 15*2           | 6.35                   | 4.115       | 15 X variable            | 11                       | 3.58                   |
|                                                                                     | 10*10*3        | 6.35                   | 4.115       | 10 x 10                  | 6 x 6                    | 5.46                   |
|                                                                                     | 12*12*3        | 6.35                   | 4.115       | 12 x 12                  | 8 x 8                    | 5.80                   |
|                                                                                     | 15*15*3        | 6.35                   | 4.115       | 15 x 15                  | 11 x 11                  | 5.25                   |
|  | 12*12*4        | 6.35                   | 4.115       | 12 x 12                  | 8 x 8                    | 6.02                   |
|                                                                                     | 15*15*4        | 6.35                   | 4.115       | 15 x 15                  | 11 x 11                  | 6.91                   |
|                                                                                     | 15*15*4*5/16"  | 7.94                   | 5.723       | 15 x 15                  | 11 x 11                  | 12.35                  |
|                                                                                     | 15*15*4 11/32  | 9.00                   | 5.723       | 15 x 15                  | 11 x 11                  | 13.00                  |
|                                                                                     | 12*20*4        | 6.35                   | 4.115       | 12 x 20                  | 8 x 16                   | 7.29                   |
|  | 15*10*4        | 6.35                   | 4.115       | 15 x 10                  | 11 x 6                   | 7.55                   |
|                                                                                     | 15*20*4        | 6.35                   | 4.115       | 15 x 20                  | 11 x 16                  | 8.34                   |
|                                                                                     | 15*25*4        | 6.35                   | 4.115       | 15 x 25                  | 11 x 21                  | 7.66                   |
|                                                                                     | 15*30*4        | 6.35                   | 4.115       | 15 x 30                  | 11 x 26                  | 8.04                   |
|                                                                                     | 15*20*4*5/16"  | 7.94                   | 5.723       | 15 x 20                  | 11 x 16                  | 12.95                  |
|                                                                                     | 15*25*4*5/16"  | 7.94                   | 5.723       | 15 x 25                  | 11 x 21                  | 13.56                  |
|                                                                                     | 15*30*4*5/16"  | 7.94                   | 5.723       | 15 x 30                  | 11 x 26                  | 14.16                  |
|                                                                                     | 15*20*4 11/32" | 9.00                   | 5.723       | 15 x 20                  | 11 x 16                  | 14.00                  |
| 15*25*4 11/32"                                                                      | 9.00           | 5.723                  | 15 x 25     | 11 x 21                  | 15.50                    |                        |
| 15*30*4 11/32"                                                                      | 9.00           | 5.723                  | 15 x 30     | 11 x 21                  | 16.00                    |                        |

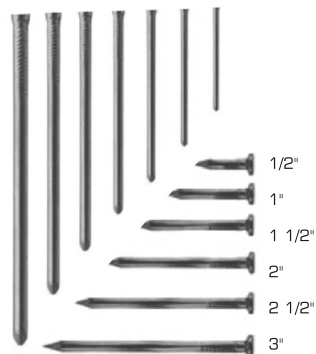
NOTA: PREGUNTE POR CASTILLO DE 3/8" Y ESCALERILLA.

## CLAVO ESTÁNDAR CON CABEZA

| Largo  |     | Calibre | Clavos por kg |
|--------|-----|---------|---------------|
| pulg   | mm  |         |               |
| 3/4"   | 20  | 14.5    | 2,362         |
| 1"     | 25  | 14.5    | 1,425         |
| 1 1/4" | 31  | 14.5    | 1,195         |
| 1 1/2" | 38  | 14.5    | 1,030         |
| 2"     | 51  | 12.5    | 440           |
| 2 1/2" | 63  | 11      | 245           |
| 3"     | 76  | 10      | 170           |
| 3 1/2" | 89  | 8       | 99            |
| 4"     | 101 | 7       | 73            |
| 5"     | 127 | 6       | 51            |
| 6"     | 152 | 5       | 35            |
| 7"     | 178 | 1       | 17            |
| 8"     | 203 | 5/16"   | 11            |

CAJA DE 25KG

CAJA CON 25 BOLSAS DE 1KG



## CLAVO PARA CONCRETO

| Largo  |       | Calibre | Clavos por kg |
|--------|-------|---------|---------------|
| pulg   | mm    |         |               |
| 1"     | 25    | 8       | 335           |
| 1 1/2" | 38    | 8       | 232           |
| 2"     | 51    | 8       | 177           |
| 2 1/2" | 63    | 8       | 143           |
| 3"     | 76    | 8       | 120           |
| 3 1/2" | 89    | 8       | 104           |
| 4"     | 101.6 | 7       | 77            |

CAJA DE 25KG

CAJA CON 25 BOLSAS DE 1KG

## CLAVO TOMATERO

| Largo  |    | Calibre | Clavos por kg |
|--------|----|---------|---------------|
| pulg   | mm |         |               |
| 1 1/4" | 32 | 14.5    | 1,095         |
| 1 3/8" | 35 | 14.5    | 1,105         |
| 1 5/8" | 41 | 14.5    | 970           |

## GRAPAS

| Calibre | Longitud | Grapas/kg | Tipo                  |
|---------|----------|-----------|-----------------------|
| 9       | 1"       | 248       | Corrugada Negra       |
| 9       | 1 1/4"   | 203       |                       |
| 8       | 1"       | 198       | Negra "V"             |
| 8       | 1 1/4"   | 172       |                       |
| 9       | 1"       | 248       | Corrugada Galvanizada |
| 9       | 1 1/4"   | 203       |                       |
| 9       | 1 1/2"   | 160       |                       |
| 10      | 1"       | 285       | Corcel III            |
| 10      | 1 1/4"   | 231       |                       |

GRAPA NEGRA CAJA CON 25KG

GRAPA GALVANIZADA CAJA CON 25KG

CAJA CON 25 BOLSAS DE 1KG

## TEE Y ZETA DE FIERRO

| Dimensiones |            | Peso |       |        |
|-------------|------------|------|-------|--------|
| pulgadas    | mm         | kg/m | largo | kg/pza |
| 1/8 x 1"    | 3.2 x 25.4 | 1.32 | 6.10  | 8.05   |

## CUADRADOS

| Dimensiones |          | Área de la Selección |       | Peso   |       |        |        |
|-------------|----------|----------------------|-------|--------|-------|--------|--------|
| mm          | pulgadas | mm                   | pulg2 | kg/m   | largo | kg/pza | lb/pie |
| 9.5         | 3/8"     | 90.73                | 0.141 | 0.712  | 6.10  | 4.34   | 0.478  |
| 12.7        | 1/2"     | 161.29               | 0.250 | 1.266  | 6.10  | 7.72   | 0.851  |
| 15.9        | 5/8"     | 252.02               | 0.399 | 1.978  | 6.10  | 12.07  | 1.329  |
| 19.1        | 3/4"     | 362.90               | 0.563 | 2.849  | 6.10  | 17.38  | 1.915  |
| 22.2        | 7/8"     | 493.95               | 0.766 | 3.877  | 6.10  | 23.65  | 2.605  |
| 25.4        | 1"       | 645.16               | 1.000 | 5.065  | 6.10  | 30.90  | 3.404  |
| 28.6        | 1 1/8"   | 816.53               | 1.266 | 6.410  | 6.10  | 39.10  | 4.308  |
| 31.8        | 1 1/4"   | 1008.07              | 1.563 | 7.913  | 6.10  | 48.27  | 5.318  |
| 34.9        | 1 3/8"   | 1219.77              | 1.891 | 9.575  | 6.10  | 58.41  | 6.434  |
| 38.1        | 1 1/2"   | 1451.62              | 2.250 | 11.395 | 6.10  | 69.51  | 7.657  |
| 44.5        | 1 3/4"   | 1975.80              | 3.065 | 15.510 | 6.10  | 94.61  | 10.423 |
| 50.8        | 2"       | 2580.65              | 4.000 | 20.258 | 6.10  | 123.57 | 13.613 |

## REDONDOS

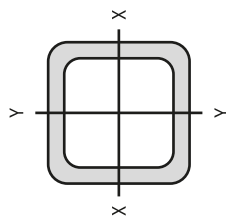
| Diámetro |          | Área de la Selección |       | Área Neta | Peso  |       |        |        |
|----------|----------|----------------------|-------|-----------|-------|-------|--------|--------|
| mm       | pulgadas | mm                   | pulg2 | cm2       | kg/m  | largo | kg/pza | lb/pie |
| 6.3      | 1/4"     | 31.7                 | 0.049 | 0.17      | 0.25  | 6.10  | 1.52   | 0.167  |
| 7.9      | 5/16"    | 49.5                 | 0.077 | 0.19      | 0.39  | 6.10  | 2.37   | 0.261  |
| 9.5      | 3/8"     | 71.3                 | 0.110 | 0.44      | 0.56  | 6.10  | 3.41   | 0.376  |
| 12.7     | 1/2"     | 126.7                | 0.196 | 0.81      | 0.99  | 6.10  | 6.06   | 0.668  |
| 15.9     | 5/8"     | 197.9                | 0.307 | 1.30      | 1.55  | 6.10  | 9.47   | 1.043  |
| 19.1     | 3/4"     | 285.0                | 0.442 | 2.01      | 2.24  | 6.10  | 13.63  | 1.502  |
| 22.2     | 7/8"     | 387.9                | 0.601 | 2.71      | 3.05  | 6.10  | 18.57  | 2.046  |
| 25.4     | 1"       | 206.7                | 0.785 | 3.55      | 3.97  | 6.10  | 24.24  | 2.669  |
| 28.6     | 1 1/8"   | 641.3                | 0.994 | 4.46      | 5.02  | 6.10  | 30.63  | 3.375  |
| 31.8     | 1 1/4"   | 791.7                | 1.227 | 5.73      | 6.21  | 6.10  | 37.87  | 4.172  |
| 34.9     | 1 3/8"   | 658.0                | 1.485 | 6.80      | 7.51  | 6.10  | 45.84  | 5.049  |
| 38.1     | 1 1/2"   | 1140.1               | 1.767 | 8.35      | 9.00  | 6.10  | 54.90  | 6.048  |
| 41.3     | 1 5/8"   | 1338.0               | 2.074 | 9.77      | 10.49 | 6.10  | 63.99  | 7.049  |
| 44.5     | 1 3/4"   | 1551.8               | 2.405 | 11.25     | 12.17 | 6.10  | 74.24  | 8.178  |
| 44.7     | 1 7/8"   | 1781.3               | 2.761 | 13.21     | 13.37 | 6.10  | 81.56  | 9.388  |
| 50.8     | 2"       | 2026.8               | 3.142 | 14.83     | 15.89 | 6.10  | 96.93  | 10.678 |

\*EL GRADO DE LOS SÓLIDOS SON A36. \*\*COTIZA CON TU EJECUTIVO DE VENTA.



## HSS CUADRADO

| Dimensiones     |  | Peso (kg/m)   |         |       |       |       |        |        |
|-----------------|--|---------------|---------|-------|-------|-------|--------|--------|
|                 |  | mm<br>(A X B) | Espesor |       |       |       |        |        |
| pulg<br>(A X B) |  |               | 1/8"    | 3/16" | 1/4"  | 5/16" | 3/8"   | 1/2"   |
| 4" X 4"         |  | 101.6 X 101.6 | 9.61    | 14.02 | 18.17 | 22.07 | 25.7   | 32.19  |
| 4 1/2" X 4 1/2" |  | 114.3 X 114.3 | 10.86   | 15.92 | 20.7  | 25.24 | 29.5   | 37.25  |
| 5" X 5"         |  | 127.0 X 127.0 | 12.13   | 17.81 | 23.25 | 28.39 | 33.29  | 42.31  |
| 5 1/2" X 5 1/2" |  | 139.7 X 139.7 |         | 19.72 | 25.77 | 31.56 | 37.1   |        |
| 6" X 6"         |  | 152.4 X 152.4 |         | 21.62 | 28.3  | 34.73 | 40.89  | 52.44  |
| 7" X 7"         |  | 177.8 X 177.8 |         | 25.42 | 33.36 | 41.06 | 48.48  | 62.58  |
| 7 1/2" X 7 1/2" |  | 190.5 X 190.5 |         | 27.31 | 35.89 | 44.23 | 52.28  | 67.64  |
| 8" X 8"         |  | 203.2 X 203.2 |         | 29.21 | 38.42 | 47.38 | 56.09  | 72.7   |
| 9" X 9"         |  | 228.6 X 228.6 |         | 33.01 | 43.5  | 53.72 | 63.68  | 82.83  |
| 10" X 10"       |  | 254.0 X 254.0 |         | 36.8  | 48.56 | 60.05 | 71.28  | 92.95  |
| 12" X 12"       |  | 304.8 X 304.8 |         | 44.41 | 58.68 | 72.71 | 86.46  | 113.2  |
| 14" X 14"       |  | 355.6 X 355.6 |         |       |       | 85.36 | 101.66 | 133.46 |
| 16" X 16"       |  | 406.4 X 406.4 |         |       |       | 98.03 | 116.85 | 153.73 |
|                 |  |               |         |       |       |       |        | 189.01 |



## HSS RECTANGULAR

| Dimensiones     |               | Peso (kg/m) |       |       |       |        |        |
|-----------------|---------------|-------------|-------|-------|-------|--------|--------|
| pulg<br>(A X B) | mm<br>(A X B) | Espesor     |       |       |       |        |        |
|                 |               | 1/8"        | 3/16" | 1/4"  | 5/16" | 3/8"   | 1/2"   |
| 5" X 3"         | 127.0 X 76.2  | 9.61        | 14.02 | 18.17 | 22.07 | 25.7   | 32.19  |
| 6" X 2"         | 152.4 X 50.8  | 9.61        | 14.02 | 18.17 | 22.07 | 25.7   | 32.19  |
| 5" X 4"         | 127.0 X 101.6 | 10.86       | 15.92 | 20.7  | 25.24 | 29.5   | 37.25  |
| 6" X 3"         | 152.4 X 76.2  | 10.86       | 15.92 | 20.7  | 25.24 | 29.5   | 37.25  |
| 7" X 2"         | 177.8 X 50.8  | 10.86       | 15.92 | 20.7  | 25.24 | 29.5   | 37.25  |
| 6" X 4"         | 152.4 X 101.6 | 12.13       | 17.81 | 23.25 | 28.39 | 33.29  | 42.31  |
| 7" X 3"         | 177.8 X 76.2  | 12.13       | 17.81 | 23.25 | 28.39 | 33.29  | 42.31  |
| 8" X 2"         | 203.2 X 50.8  | 12.13       | 17.81 | 23.25 | 28.39 | 33.29  | 42.31  |
| 6" X 5"         | 152.4 X 127.0 |             | 19.72 | 25.77 | 31.56 | 37.1   | 52.44  |
| 7" X 4"         | 177.8 X 101.6 |             | 19.72 | 25.77 | 31.56 | 37.1   | 52.44  |
| 8" X 3"         | 203.2 X 76.2  |             | 19.72 | 25.77 | 31.56 | 37.1   | 52.44  |
| 7" X 5"         | 177.8 X 127.0 |             | 21.62 | 28.3  | 34.73 | 40.89  | 52.44  |
| 8" X 4"         | 203.2 X 101.6 |             | 21.62 | 28.3  | 34.73 | 40.89  | 62.58  |
| 9" X 3"         | 228.6 X 76.2  |             | 21.62 | 28.3  | 34.73 | 40.89  | 62.58  |
| 10" X 2"        | 254.0 X 50.8  |             | 21.62 | 28.3  | 34.73 | 40.89  | 62.58  |
| 8" X 6"         | 203.2 X 152.4 |             | 25.42 | 33.36 | 41.06 | 48.48  | 62.58  |
| 9" X 5"         | 228.6 X 127.0 |             | 25.42 | 33.36 | 41.06 | 48.48  | 67.64  |
| 10" X 4"        | 254.0 X 101.6 |             | 25.42 | 33.36 | 41.06 | 48.48  | 67.64  |
| 12" X 2"        | 304.8 X 50.8  |             | 25.42 | 33.36 | 41.06 | 48.48  | 67.64  |
| 8" X 7"         | 203.2 X 177.8 |             | 27.31 | 35.89 | 44.23 | 52.28  | 67.64  |
| 9" X 6"         | 228.6 X 152.4 |             | 27.31 | 35.89 | 44.23 | 52.28  | 72.7   |
| 10" X 5"        | 254.0 X 127.0 |             | 27.31 | 35.89 | 44.23 | 52.28  | 72.7   |
| 12" X 3"        | 304.8 X 76.2  |             | 27.31 | 35.89 | 44.23 | 52.28  | 72.7   |
| 9" X 7"         | 228.6 X 177.8 |             | 29.21 | 38.42 | 47.38 | 56.09  | 82.83  |
| 10" X 6"        | 254.0 X 152.4 |             | 29.21 | 38.42 | 47.38 | 56.09  | 72.7   |
| 12" X 4"        | 304.8 X 101.6 |             | 29.21 | 38.42 | 47.38 | 56.09  | 82.83  |
| 10" X 8"        | 254.0 X 203.2 |             | 33.01 | 43.5  | 53.72 | 63.68  | 92.95  |
| 12" X 6"        | 304.8 X 152.4 |             | 33.01 | 43.5  | 53.72 | 63.68  | 92.95  |
| 14" X 4"        | 355.6 X 101.6 |             | 33.01 | 43.5  | 53.72 | 63.68  | 92.95  |
| 12" X 8"        | 304.8 X 203.2 |             | 36.8  | 48.56 | 60.05 | 71.28  | 113.2  |
| 14" X 6"        | 355.6 X 152.4 |             | 36.8  | 48.56 | 60.05 | 71.28  | 113.2  |
| 16" X 4"        | 406.4 X 101.6 |             | 36.8  | 48.56 | 60.05 | 71.28  | 113.2  |
| 14" X 10"       | 355.6 X 254.0 |             | 44.41 | 58.68 | 72.71 | 86.46  | 113.2  |
| 16" X 8"        | 406.4 X 203.2 |             | 44.41 | 58.68 | 72.71 | 86.46  | 133.46 |
| 18" X 6"        | 457.2 X 152.4 |             | 44.41 | 58.68 | 72.71 | 86.46  | 133.46 |
| 20" X 4"        | 508.0 X 101.6 |             | 44.41 | 58.68 | 72.71 | 86.46  | 153.73 |
| 16" x 12"       | 406.4 X 304.8 |             |       |       | 85.36 | 101.66 |        |
| 20" x 8"        | 508.0 X 203.2 |             |       |       | 85.36 | 101.66 |        |
| 20" x 12"       | 508.0 X 304.8 |             |       |       | 98.03 | 116.85 |        |

## LÁMINA GALVANIZADA

| Pérfil    | Calibre | Peso<br>kg/m <sup>2</sup> | Compresión superior<br>momento positivo |              |                                                 | Compresión superior<br>momento negativo                                                          |                         |               | Acciones permisibles     |                    |      |
|-----------|---------|---------------------------|-----------------------------------------|--------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------|---------------|--------------------------|--------------------|------|
|           |         |                           | I+<br>cm4/mt                            | S+<br>cm3/mt | MMAx+<br>kg-m                                   | I-<br>cm4/mt                                                                                     | S-<br>cm3/mt            | MMAx-<br>kg-m | Valor<br>máximo<br>kg/mt | Reacciones máximas |      |
|           |         |                           |                                         |              |                                                 |                                                                                                  |                         |               |                          |                    |      |
| RN-100/35 | 26      | 4.69                      | 10.57                                   | 4.42         | 68.95                                           | 6.86                                                                                             |                         | 57.41         | 1065                     | 161                | 201  |
|           | 24      | 5.42                      | 12.76                                   | 5.38         | 83.93                                           | 8.21                                                                                             |                         | 69.73         | 1452                     | 208                | 317  |
|           | 22      | 7.60                      | 18.53                                   | 8.05         | 125.58                                          | 12.53                                                                                            |                         | 110.92        | 2449                     | 379                | 772  |
| R-72      | 30      | 2.44                      | 2.90                                    | 1.73         | 26.98                                           | 2.02                                                                                             |                         | 24.34         | 742                      | 147                | 181  |
|           | 28      | 2.97                      | 3.93                                    | 2.41         | 37.60                                           | 2.66                                                                                             |                         | 32.92         | 1279                     | 208                | 330  |
|           | 26      | 3.52                      | 4.93                                    | 3.07         | 47.89                                           | 3.38                                                                                             |                         | 42.59         | 1846                     | 279                | 515  |
|           | 24      | 4.06                      | 5.81                                    | 3.65         | 56.94                                           | 4.14                                                                                             |                         | 50.70         | 2372                     | 359                | 731  |
|           | 22      | 5.70                      | 8.33                                    | 5.34         | 83.30                                           | 6.78                                                                                             |                         | 76.44         | 3393                     | 680                | 1157 |
| R-101     | 28      | 3.46                      | 3.93                                    | 2.41         | 37.60                                           | 2.66                                                                                             |                         | 32.92         | 1279                     | 208                | 330  |
|           | 26      | 4.69                      | 4.93                                    | 3.07         | 47.89                                           | 3.38                                                                                             |                         | 42.59         | 1846                     | 279                | 515  |
|           | 24      | 5.42                      | 5.81                                    | 3.65         | 56.94                                           | 4.14                                                                                             |                         | 50.70         | 2372                     | 359                | 731  |
|           | 22      | 7.60                      | 8.33                                    | 5.34         | 83.30                                           | 6.78                                                                                             |                         | 73.44         | 3393                     | 680                | 1557 |
| O-725     | 30      | 2.44                      | 1.84                                    | 1.90         | 29.64                                           | ancho efectivo 71.46 cm<br>ancho total 79.71 cm                                                  |                         |               |                          |                    |      |
|           | 28      | 2.97                      | 2.28                                    | 2.35         | 36.68                                           |                                                                                                  |                         |               |                          |                    |      |
|           | 26      | 3.52                      | 2.74                                    | 2.82         | 44.00                                           |                                                                                                  |                         |               |                          |                    |      |
|           | 24      | 4.06                      | 3.20                                    | 3.30         | 51.50                                           |                                                                                                  |                         |               |                          |                    |      |
| O-100     | 28      | 3.96                      | 2.28                                    | 2.35         | 36.66                                           | ancho efectivo 98.28 cm (cubiertas)<br>ancho efectivo 103.22 cm (muros)<br>ancho total 106.36 cm |                         |               |                          |                    |      |
|           | 26      | 4.69                      | 2.74                                    | 2.82         | 44.00                                           |                                                                                                  |                         |               |                          |                    |      |
|           | 24      | 5.42                      | 3.20                                    | 3.30         | 51.50                                           |                                                                                                  |                         |               |                          |                    |      |
| SECC-25   | 24      | 5.42                      | 45.11                                   | 11.93        |                                                 | 50.54                                                                                            | ancho efectivo 91.44 cm |               |                          |                    |      |
|           | 22      | 7.60                      | 69.54                                   | 19.22        |                                                 | 72.31                                                                                            |                         |               |                          |                    |      |
|           | 20      | 9.06                      | 86.34                                   | 24.54        |                                                 | 86.81                                                                                            |                         |               |                          |                    |      |
|           | 18      | 11.96                     | 114.63                                  | 35.25        |                                                 | 114.63                                                                                           |                         |               |                          |                    |      |
| R-90      | 26      | 5.21                      | 81.48                                   | 8.92         | ancho efectivo 90.00 cm<br>ancho total 95.55 cm |                                                                                                  |                         |               |                          |                    |      |
|           | 24      | 6.02                      | 100.07                                  | 11.26        |                                                 |                                                                                                  |                         |               |                          |                    |      |
|           | 22      | 8.44                      | 155.93                                  | 18.39        |                                                 |                                                                                                  |                         |               |                          |                    |      |

### NOTA:

FABRICAMOS LÁMINA RN-100/35, R-72, R-101, SECC-25 A MEDIDAS ESPECIALES

## LÁMINA NEGRA

| Espesor |        |        | Peso en kilogramos |         |          |         |          |        |                    |
|---------|--------|--------|--------------------|---------|----------|---------|----------|--------|--------------------|
| Cálbre  | pulg   | mm     | 3" X 6"            | 3" X 8" | 3" X 10" | 4" X 8" | 4" X 10" | kg/m2  |                    |
| 3       | 0.2391 | 6.0731 | 79.699             | 106.265 | 132.832  | 141.687 | 177.109  | 47.660 | Rolada en caliente |
| 4       | 0.2242 | 5.6947 | 74.732             | 99.643  | 124.554  | 132.858 | 166.072  | 44.690 |                    |
| 5       | 0.2092 | 5.3137 | 69.732             | 92.977  | 116.221  | 123.969 | 154.961  | 41.700 |                    |
| 6       | 0.1943 | 4.9352 | 64.766             | 86.354  | 107.934  | 115.139 | 143.924  | 38.730 |                    |
| 7       | 0.1793 | 4.5542 | 59.766             | 79.688  | 99.610   | 106.251 | 132.813  | 35.740 |                    |
| 8       | 0.1644 | 4.1758 | 54.799             | 73.066  | 91.332   | 97.421  | 121.776  | 32.770 |                    |
| 9       | 0.1495 | 3.7973 | 49.833             | 66.444  | 83.055   | 88.591  | 110.739  | 29.800 |                    |
| 10      | 0.1345 | 3.4163 | 44.833             | 59.777  | 74.721   | 79.703  | 99.628   | 26.810 |                    |
| 1/8"    | 0.1250 | 3.1750 | 41.666             | 55.555  | 69.444   | 74.073  | 92.591   | 24.916 |                    |
| 11      | 0.1196 | 3.0378 | 39.866             | 53.155  | 66.444   | 70.878  | 88.591   | 23.840 | LCR ó LRF          |
| 12      | 0.1046 | 2.6568 | 34.866             | 46.488  | 58.110   | 61.984  | 77.481   | 20.850 |                    |
| 13      | 0.0897 | 2.2784 | 29.900             | 39.866  | 49.833   | 53.155  | 66.444   | 17.880 |                    |
| 14      | 0.0747 | 1.8974 | 24.900             | 33.200  | 41.499   | 44.266  | 55.333   | 14.890 |                    |
| 15      | 0.0673 | 1.7094 | 22.433             | 29.911  | 37.388   | 39.881  | 49.851   | 13.415 |                    |
| 16      | 0.0598 | 1.5189 | 19.933             | 26.577  | 33.222   | 35.437  | 44.296   | 11.920 |                    |
| 17      | 0.0538 | 1.3665 | 17.933             | 23.911  | 29.889   | 31.881  | 39.851   | 10.724 |                    |
| 18      | 0.0478 | 1.2141 | 15.933             | 21.244  | 26.555   | 28.326  | 35.407   | 9.528  |                    |
| 19      | 0.0418 | 1.0617 | 13.933             | 18.578  | 23.222   | 24.770  | 30.963   | 8.332  |                    |
| 20      | 0.0359 | 0.9119 | 11.967             | 15.955  | 19.944   | 21.274  | 26.592   | 7.156  | Rolada en frío     |
| 21      | 0.0329 | 0.8357 | 10.967             | 14.622  | 18.278   | 19.496  | 24.370   | 6.558  |                    |
| 22      | 0.0299 | 0.7595 | 9.967              | 13.289  | 16.611   | 17.718  | 22.148   | 5.960  |                    |
| 23      | 0.0269 | 0.6833 | 8.967              | 11.955  | 14.944   | 15.941  | 19.926   | 5.362  |                    |
| 24      | 0.0239 | 0.6071 | 7.967              | 10.622  | 13.278   | 14.163  | 17.703   | 4.764  |                    |
| 25      | 0.0209 | 0.5309 | 6.967              | 9.289   | 11.611   | 12.385  | 15.481   | 4.166  |                    |
| 26      | 0.0179 | 0.4547 | 5.967              | 7.955   | 9.944    | 10.607  | 13.259   | 3.568  |                    |
| 27      | 0.0164 | 0.4166 | 5.467              | 7.289   | 9.111    | 9.718   | 12.148   | 3.269  |                    |
| 28      | 0.0149 | 0.3785 | 4.967              | 6.622   | 8.278    | 8.830   | 11.037   | 2.970  |                    |
| 29      | 0.0135 | 0.3429 | 4.500              | 6.000   | 7.500    | 8.000   | 10.000   | 2.691  |                    |
| 30      | 0.0120 | 0.3048 | 4.000              | 5.333   | 6.667    | 7.111   | 8.889    | 2.392  |                    |

## LÁMINA Y PLACA ANTIDERRAPANTE

| Espesor |       | Pesos teóricos |         |          |          |
|---------|-------|----------------|---------|----------|----------|
| Cálbre  | pulg  | 3" X 8"        | 4" X 8" | 3" X 10" | 4" X 10" |
| 1/4"    | 0.250 | 122.58         | 163.44  | 145.00   | 195.00   |
| 3/16"   | 0.188 | 94.82          | 126.43  | 119.00   | 155.00   |
| 10      | 0.134 | 74.13          | 98.84   | 79.50    | 105.00   |
| 1/8"    | 0.125 | 67.06          | 89.42   |          |          |
| 11      | 0.119 | 64.57          | 86.09   |          |          |
| 12      | 0.104 | 57.14          | 76.19   | 63.00    | 86.00    |
| 13      | 0.089 | 48.98          | 65.31   |          |          |
| 14      | 0.074 | 40.83          | 54.43   | 49.00    | 64.50    |

\*LOS PESOS PUEDEN VARIAR VS EL ESPESOR.

## CARACTERÍSTICAS DE LA MALLA ELECTROSOLDADA DE FABRICACIÓN ESTÁNDAR

| Tipo      | Diámetro del alambre (mm) | Área de acero (cm <sup>2</sup> /m) | Peso (kg/m <sup>2</sup> ) | Peso (kg/hoja) | Tipo de rollo |
|-----------|---------------------------|------------------------------------|---------------------------|----------------|---------------|
| 6x6 2/2   | 6.67                      | 2.292                              | 3.64                      | 54.59          | *             |
| 6x4 3/3   | 6.19                      | 1.971                              | 3.14                      | 47.11          | *             |
| 6x6 4/4   | 5.72                      | 1.686                              | 2.72                      | 40.81          | (A)           |
| 6x6 6/6   | 4.88                      | 1.227                              | 1.98                      | 26.69          | (A)           |
| 6x6 8/8   | 4.11                      | 0.871                              | 1.41                      | 21.01          | (A)           |
| 6x6 10/10 | 3.43                      | 0.606                              | 0.97                      | 14.54          | (A)           |
| 8x8 8/8   | 4.12                      | 0.665                              | 1.06                      | *              | (A)           |

## ALAMBRE GALVANIZADO FERRETERO

| Tipo         | Calibre | Diámetro (mm) | Tensión                                         | Peso (kg/rollo)      |
|--------------|---------|---------------|-------------------------------------------------|----------------------|
| GALVANIZADOS | 10.5    | 3.15          | 55,00 - 75,000 lb/pulg2<br>3,900 - 5,200 kg/cm2 | 25 kg<br>y<br>100 kg |
|              | 12.5    | 2.43          |                                                 |                      |
|              | 14.5    | 1.85          |                                                 |                      |
|              | 16      | 1.57          |                                                 |                      |
| PVC          | 10.5    | 3.25          | 55,00 - 75,000 lb/pulg2<br>3,900 - 5,200 kg/cm2 | 25 kg<br>y<br>100 kg |

## PLAFÓN

| Aberturas (mm) | Ancho (mt) | Calibre de la lámina | Diseño         | Largo (mt) |
|----------------|------------|----------------------|----------------|------------|
| 5 X 11         | 0.91       | 26                   | 500, 600 Y 700 | 22         |
| 8 X 25         | 0.91       | 26                   | 500            | 22         |

## MAILLA MOSQUITERA LIGERA

| Aberturas X pulg2 | Altura (cm) | Diámetro del alambre (mm) | Longitud rollos (mt) |
|-------------------|-------------|---------------------------|----------------------|
| 60                | 0.23        | 30                        | 18 X 14              |
| 75                | 0.23        | 30                        | 18 X 14              |
| 90                | 0.23        | 30                        | 18 X 14              |
| 105               | 0.23        | 30                        | 18 X 14              |
| 120               | 0.23        | 30                        | 18 X 14              |

## MAILLA MOSQUITERA GALVANIZADA REFORZADA

| Aberturas X pulg2 | Altura (cm) | Diámetro del alambre (mm) | Longitud rollos (mt) |
|-------------------|-------------|---------------------------|----------------------|
| 60                | 0.28        | 10 y 30                   | 18 X 14              |
| 75                | 0.28        | 10 y 30                   | 18 X 14              |
| 90                | 0.28        | 10 y 30                   | 18 X 14              |
| 105               | 0.28        | 10 y 30                   | 18 X 14              |
| 120               | 0.28        | 10 y 30                   | 18 X 14              |

## PANEL ESTRUCTURAL

| Espesor | Largo | Ancho | Color    | Uso          |
|---------|-------|-------|----------|--------------|
| 2"      | 1.22  | 2.44  | Blanco   | Muro         |
| 3"      | 1.22  | 2.44  | Blanco   | Muro         |
| 2"      | 1.22  | 2.44  | Amarillo | Muro / Techo |
| 3"      | 1.22  | 2.44  | Amarillo | Muro / Techo |
| 4"      | 1.22  | 2.44  |          |              |

## CRIBA DE GRANO DE PLATA LIGERA

| Medida Comercial | Claro abertura (mm) | Diámetro del alambre (mm) | Peso apro. por rollo 30 mt (kg) | Kg Zinc por rollo 30 mt | Largo (mt) | Ancho (mt) |
|------------------|---------------------|---------------------------|---------------------------------|-------------------------|------------|------------|
| 2 X 2            | 11.7                | 0.94                      | 29.1                            | 8.6                     | 30         | 0.91       |
| 3 X 3            | 7.7                 | 0.7                       | 25.1                            | 8.4                     | 30         | 0.91       |
| 4 X 4            | 5.7                 | 0.55                      | 23.6                            | 9.4                     | 30         | 0.91       |
| 5 X 5            | 4.5                 | 0.5                       | 23.1                            | 8.8                     | 30         | 0.91       |
| 6 X 6            | 3.7                 | 0.44                      | 20.5                            | 8                       | 30         | 0.91       |
| 8 X 8            | 2.7                 | 0.37                      | 23.1                            | 9.5                     | 30         | 0.91       |

## MALLA DE PÁJARO

| Aberturas (mm) | Altura (mt)      | Calibre | Largo (mt) |
|----------------|------------------|---------|------------|
| 13             | 0.9              | 23      | 20 y 45    |
| 20             | 1.20, 1.50, 1.75 | 22      | 45         |
| 25             | 1.00, 1.50, 1.75 | 22      | 45         |
| 38             | 1.50, 1.75       | 20      | 45         |
| 50             | 1.50, 1.75       | 20      | 45         |

## MALLA BORREGUERA

| Altura (mm) | Calibre alambre superior e inferior | Calibre alambre intermedio | Abertura (cm) |
|-------------|-------------------------------------|----------------------------|---------------|
| 1           | 11                                  | 14.5                       | 15            |
| 1.2         | 11                                  | 14.5                       | 15            |
| 1.4         | 11                                  | 14.5                       | 15            |
| 1.67        | 11                                  | 14.5                       | 15            |
| 2           | 11                                  | 14.5                       | 15            |
| 2.5         | 11                                  | 14.5                       | 15            |

## MULTIMALLA FERRETERA

| Abertura (cm) | Calibre | Altura (mt) | Largo (mt) | Acabado     |
|---------------|---------|-------------|------------|-------------|
| 5X5, 5X10     | 12.5    | 1.00, 1.22  | 20         | Galvanizada |
| 5X5, 5X10     | 14      | 1.00, 1.22  | 20         | Galvanizada |
| 7.5           | 10.5    | 1.22        | 20         | Galvanizada |
| 5X5           | 10.5    | 1.22        | 20         | Galvanizada |
| 2.5X5         | 14      | 1.00        | 20         | Galvanizada |
| 2.5X5         | 14      | 1.22        | 20         | Galvanizada |

## MALLA CRIBA ORNAMENTAL GALVANIZADA

| Abertura (mm) | Altura (mt) | Calibre | Largo (mt) | Acabado     |
|---------------|-------------|---------|------------|-------------|
| 50 X 50       | 1           | 10.5    | 10         | Galvanizada |
| 25 X 25       | 1           | 10.5    | 10         | Galvanizada |
| 38 X 38       | 1           | 10.5    | 10         | Galvanizada |

## CONCERTINA COMERCIAL

| Peso x rollo | Rendimiento x rollo | Calibre | Diámetro |
|--------------|---------------------|---------|----------|
| 3 kg         | 8 mt                | 24      | 18"      |

\* FABRICADA ESPECIALMENTE PARA LA PROTECCIÓN EN BARDAS O CERCAS. SE UTILIZA EN RESIDENCIAS, DEPORTIVOS, AEROPUERTOS, PROTECCIÓN DE GANADO, CENTROS DE READAPTACIÓN, HOSPITALES, ESCUELAS, BODEGAS, ETC.

\* ES UN PRODUCTO GALVANIZADO EN ROLLOS CON CUCHILLAS TIPO ARPÓN CON FILOS DE BISTURÍ.

\* PARA SU FÁCIL MANEJO Y PROTECCIÓN SE EMPACA EN CAJAS DE CARTÓN.

| Descripción |    | Cal  | Peso  |       | Dimensión |           |         | Área<br>(cm2) | Eje X - X |           |          | Eje Y - Y |           |         | Factor<br>Columna |
|-------------|----|------|-------|-------|-----------|-----------|---------|---------------|-----------|-----------|----------|-----------|-----------|---------|-------------------|
|             |    |      | kg/ml | kg/pz | d<br>mm   | t<br>pulg | R<br>mm |               | lx<br>cm4 | Sx<br>cm3 | rx<br>cm | ly<br>cm4 | Sy<br>cm3 | r<br>cm |                   |
| D X B       |    |      |       |       |           |           |         |               |           |           |          |           |           |         | Q                 |
| 3" X 1 1/2" | 12 | 3.60 | 21.60 | 19    | 2.66      | 9.5       | 4.54    | 38.5          | 10.1      | 2.9       | 4.16     | 9.56      | 1.45      | 1.51    | *                 |
|             | 14 | 2.40 | 14.40 | 19    | 0.0747    | 6.4       | 3.23    | 28            | 7.22      | 2.92      | 7        | 2.99      | 1.46      | 14.99   | 0.98              |
|             | 16 | 1.99 | 11.94 | 19    | 1.52      | 9.5       |         |               |           |           |          |           |           |         |                   |
|             |    |      |       |       |           |           |         |               |           |           |          |           |           |         |                   |
| 4" X 2"     | 10 | 5.85 | 23.40 | 19    | 0.1345    | 6.4       | 7.30    | 113           | 22.15     | 3.93      | 25       | 7.66      | 1.84      | 18.35   | 1.00              |
|             | 12 | 4.55 | 18.20 | 19    | 0.1046    | 6.4       | 5.78    | 91            | 17.86     | 3.96      | 20       | 6.3       | 1.88      | 18.36   | 0.97              |
|             | 14 | 3.29 | 13.16 | 19    | 0.0747    | 6.4       | 4.19    | 67            | 13.22     | 4.00      | 15       | 4.76      | 1.92      | 18.37   | 0.91              |
|             | 16 | 2.72 | 16.32 | 19    | 1.52      | 9.5       |         |               |           |           |          |           |           | 1.82    |                   |
| 5" X 2"     | 10 | 6.56 | 32.80 | 19    | 0.1345    | 6.4       | 8.17    | 192           | 30.25     | 4.85      | 27       | 7.89      | 1.82      | 16.58   | 0.97              |
|             | 12 | 5.08 | 25.40 | 19    | 0.1046    | 6.4       | 6.45    | 154           | 24.29     | 4.89      | 22       | 6.49      | 1.86      | 16.58   | 0.92              |
|             | 14 | 3.68 | 18.40 | 19    | 0.0747    | 6.4       | 4.68    | 114           | 17.90     | 4.93      | 17       | 4.89      | 1.89      | 16.57   | 0.84              |
|             |    |      |       |       |           |           |         |               |           |           |          |           |           |         |                   |
| 6" X 2"     | 10 | 7.38 | 44.28 | 19    | 0.1345    | 6.4       | 9.04    | 298           | 39.12     | 5.74      | 29       | 8.06      | 1.78      | 15.15   | 0.92              |
|             | 12 | 5.82 | 34.92 | 19    | 0.1046    | 6.4       | 7.13    | 239           | 31.31     | 5.79      | 24       | 6.62      | 1.82      | 15.13   | 0.85              |
|             | 14 | 4.21 | 25.26 | 19    | 0.0747    | 6.4       | 5.16    | 175           | 23.00     | 5.83      | 18       | 4.99      | 1.86      | 15.11   | 0.77              |
|             |    |      |       |       |           |           |         |               |           |           |          |           |           |         |                   |
| 6" X 2 1/2" | 10 | 7.95 | 47.70 | 19    | 0.1345    | 6.4       | 9.90    | 346           | 45.44     | 5.91      | 50       | 11.54     | 2.25      | 19.89   | 0.93              |
|             | 12 | 6.18 | 37.08 | 19    | 0.1046    | 6.4       | 7.80    | 276           | 36.27     | 5.95      | 41       | 9.39      | 2.29      | 19.90   | 0.87              |
|             | 14 | 4.46 | 26.76 | 19    | 0.0747    | 6.4       | 5.64    | 203           | 26.54     | 5.99      | 31       | 7.02      | 2.33      | 19.90   | 0.79              |
|             |    |      |       |       |           |           |         |               |           |           |          |           |           |         |                   |
| 7" X 2 3/4" | 10 | 9.00 | 63.00 | 22    | 0.1345    | 6.4       | 11.41   | 542           | 60.94     | 6.89      | 72       | 14.89     | 2.51      | 21.62   | 0.88              |
|             | 12 | 6.99 | 48.93 | 19    | 0.1046    | 6.4       | 8.81    | 424           | 47.64     | 6.93      | 55       | 11.12     | 2.49      | 20.76   | 0.82              |
|             | 14 | 5.04 | 35.28 | 19    | 0.0747    | 6.4       | 6.36    | 309           | 34.37     | 6.97      | 41       | 8.27      | 2.53      | 20.76   | 0.74              |
|             |    |      |       |       |           |           |         |               |           |           |          |           |           |         |                   |
| 8" X 2 3/4" | 10 | 9.82 | 78.56 | 22    | 0.1345    | 6.4       | 12.28   | 744           | 73.19     | 7.78      | 75       | 15.11     | 2.47      | 20.21   | 0.83              |
|             | 12 | 7.72 | 61.76 | 19    | 0.1046    | 6.4       | 9.49    | 580           | 57.10     | 7.82      | 57       | 11.28     | 2.45      | 19.38   | 0.77              |
|             | 14 | 5.57 | 44.56 | 19    | 0.0747    | 6.4       | 6.85    | 423           | 41.16     | 7.86      | 42       | 8.39      | 2.49      | 19.37   | 0.69              |
|             |    |      |       |       |           |           |         |               |           |           |          |           |           |         |                   |

FABRICAMOS POLÍN A MEDIDAS ESPECIALES.

NOTAS:

\* PESOS TEÓRICOS CALCULADOS SEGÚN PENALTE DEL POLÍN.

\* POLÍN DE 3" CALCULADO A 6 MTS.

\* POLÍN DE 4" SE SUJETA TAMBIÉN EN 4, 5 Y 6 MTS.

\* POLÍN DE 5" SE SUJETA TAMBIÉN EN 5 Y 6 MTS.

POLÍN MON-TEN

| Descripción |              | Cal | Peso  |        | Dimensión |        | Área<br>(cm2) | Eje X - X |      |        | Eje Y - Y |     |       | Factor<br>Columna |       |      |
|-------------|--------------|-----|-------|--------|-----------|--------|---------------|-----------|------|--------|-----------|-----|-------|-------------------|-------|------|
|             |              |     | kg/ml | kg/pz  | d         | t      |               | R         | lx   | Sx     | rx        | ly  | Sy    | r                 | x     |      |
| D X B       |              |     |       |        | mm        | pulg   | mm            | cm4       | cm3  | cm     | cm4       | cm3 | cm    | mm                | g     |      |
|             | 8" X 3"      | 10  | 10.14 | 81.12  | 22        | 0.1345 | 6.4           | 12.71     | 787  | 77.45  | 7.87      | 93  | 17.34 | 2.71              | 22.53 | 0.84 |
|             | 8" X 3"      | 12  | 7.89  | 63.12  | 22        | 0.1046 | 6.4           | 9.98      | 625  | 61.47  | 7.91      | 75  | 14.00 | 2.74              | 22.52 | 0.78 |
|             | 8" X 3"      | 14  | 5.63  | 45.04  | 19        | 0.0747 | 6.4           | 7.09      | 448  | 42.80  | 7.95      | 52  | 9.62  | 2.72              | 21.67 | 0.69 |
|             | 9" X 2 3/4"  | 10  | 10.52 | 94.68  | 22        | 0.1345 | 6.4           | 13.15     | 985  | 86.18  | 8.66      | 78  | 15.29 | 2.43              | 18.99 | 0.79 |
|             | 9" X 2 3/4"  | 12  | 8.26  | 74.34  | 19        | 0.1046 | 6.4           | 10.16     | 767  | 67.14  | 8.69      | 59  | 11.42 | 2.41              | 18.18 | 0.72 |
|             | 9" X 2 3/4"  | 14  | 5.96  | 53.64  | 19        | 0.0747 | 6.4           | 7.33      | 559  | 48.36  | 8.73      | 44  | 8.49  | 2.45              | 18.16 | 0.65 |
|             | 9" X 3 1/2"  | 10  | 11.78 | 106.02 | 25        | 0.1345 | 6.4           | 14.65     | 1167 | 102.10 | 8.92      | 150 | 24.02 | 3.19              | 26.65 | 0.81 |
|             | 9" X 3 1/2"  | 12  | 9.12  | 82.08  | 22        | 0.1046 | 6.4           | 11.33     | 911  | 79.09  | 8.96      | 114 | 18.07 | 3.17              | 25.79 | 0.75 |
|             | 9" X 3 1/2"  | 14  | 6.56  | 59.04  | 19        | 0.0747 | 6.4           | 8.05      | 652  | 53.19  | 9.00      | 80  | 12.45 | 3.15              | 24.91 | 0.63 |
|             | 10" X 2 3/4" | 10  | 11.22 | 112.20 | 22        | 0.1345 | 6.4           | 14.01     | 1269 | 99.92  | 9.52      | 80  | 15.44 | 2.39              | 17.92 | 0.75 |
|             | 10" X 2 3/4" | 12  | 8.80  | 88.00  | 19        | 0.1046 | 6.4           | 10.84     | 987  | 77.75  | 9.55      | 61  | 11.53 | 2.37              | 17.13 | 0.68 |
|             | 10" X 2 3/4" | 14  | 6.34  | 63.40  | 19        | 0.0747 | 6.4           | 7.81      | 719  | 55.97  | 9.59      | 45  | 8.58  | 2.41              | 17.09 | 0.61 |
|             | 10" X 3 1/2" | 10  | 12.37 | 123.70 | 25        | 0.1345 | 6.4           | 15.52     | 1495 | 117.74 | 9.82      | 155 | 24.29 | 3.16              | 25.26 | 0.77 |
|             | 10" X 3 1/2" | 12  | 9.62  | 96.20  | 22        | 0.1046 | 6.4           | 12.01     | 1165 | 91.12  | 9.85      | 118 | 18.28 | 3.13              | 24.42 | 0.71 |
|             | 10" X 3 1/2" | 14  | 6.87  | 68.70  | 19        | 0.0747 | 6.4           | 8.53      | 833  | 61.37  | 9.88      | 82  | 12.59 | 3.11              | 23.55 | 0.60 |
|             | 12" X 3 1/2" | 10  | 13.74 | 164.88 | 25        | 0.1345 | 6.4           | 17.26     | 2308 | 151.37 | 11.56     | 163 | 24.73 | 3.08              | 22.88 | 0.71 |
|             | 12" X 3 1/2" | 12  | 10.64 | 127.68 | 22        | 0.1046 | 6.4           | 13.36     | 1796 | 117.02 | 11.59     | 124 | 18.61 | 3.05              | 22.07 | 0.64 |
|             | 12" X 3 1/2" | 14  | 7.57  | 90.84  | 19        | 0.0747 | 6.4           | 9.50      | 1282 | 79.06  | 11.62     | 87  | 12.82 | 3.02              | 21.25 | 0.54 |

FABRICAMOS POLÍN A MEDIDAS ESPECIALES.

NOTAS:

\* PESOS TEÓRICOS CALCULADOS SEGÚN PERALTE DEL POLÍN.

\* POLÍN DE 3" CALCULADO A 6 MTS.

\* POLÍN DE 4" SE SURTE TAMBIÉN EN 4, 5 Y 6 MTS.

\* POLÍN DE 5" SE SURTE TAMBIÉN EN 5 Y 6 MTS.



## PLACA

| Dimensiones |              | Peso (kg) Espesor |        |        |      |        |        |        |        |        |        |      |      |      |        |      |       |      |       |      |       |      |       |  |
|-------------|--------------|-------------------|--------|--------|------|--------|--------|--------|--------|--------|--------|------|------|------|--------|------|-------|------|-------|------|-------|------|-------|--|
|             |              | 3"                | 2 1/2" | 2 1/4" | 2"   | 1 3/4" | 1 5/8" | 1 1/2" | 1 3/8" | 1 1/4" | 1 1/8" | 1"   | 7/8" | 3/4" | 11/16" | 5/8" | 9/16" | 1/2" | 7/16" | 3/8" | 5/16" | 1/4" | 3/16" |  |
| pies        | metros       | mm                | mm     | mm     | mm   | mm     | mm     | mm     | mm     | mm     | mm     | mm   | mm   | mm   | mm     | mm   | mm    | mm   | mm    | mm   | mm    | mm   | mm    |  |
| 3x10        | 0.914 x 3.05 |                   |        |        |      |        |        |        |        |        |        |      |      |      |        |      |       |      |       | 211  | 176   | 141  | 106   |  |
| 3x20        | 0.914 x 6.10 |                   |        |        |      |        |        |        |        |        |        |      |      |      |        |      |       |      |       | 421  | 351   | 281  | 211   |  |
| 4x10        | 1.22 x 3.05  |                   |        |        |      |        |        |        |        |        |        |      |      |      |        |      |       |      |       | 281  | 234   | 187  | 141   |  |
| 4x20        | 1.22 x 6.10  |                   |        |        |      |        |        |        |        |        |        |      |      |      |        |      |       |      |       | 562  | 468   | 375  | 281   |  |
| 5x10        | 1.52 x 3.05  | 2809              | 2341   | 2107   | 1873 | 1638   | 1521   | 1404   | 1287   | 1170   | 1053   | 936  | 819  | 702  | 643    | 585  | 526   | 468  | 409   | 351  | 293   | 234  | 176   |  |
| 5x20        | 1.52 x 6.10  | 5617              | 4681   | 4213   | 3745 | 3276   | 3042   | 2808   | 2574   | 2340   | 2106   | 1872 | 1638 | 1404 | 1286   | 1170 | 1052  | 936  | 818   | 702  | 585   | 4686 | 352   |  |
| 6x10        | 1.83 x 3.05  | 3370              | 2809   | 2528   | 2247 | 1966   | 1826   | 1685   | 1545   | 1404   | 1264   | 1124 | 983  | 843  | 772    | 702  | 632   | 562  | 491   | 421  | 351   | 281  | 211   |  |
| 6x20        | 1.83 x 6.10  | 6740              | 5617   | 5055   | 4494 | 3932   | 3651   | 3370   | 3089   | 2808   | 2528   | 2247 | 1966 | 1685 | 1543   | 1404 | 1263  | 1123 | 982   | 842  | 702   | 562  | 422   |  |
| 7x10        | 2.14 x 3.05  | 3932              | 3277   | 2949   | 2621 | 2294   | 2130   | 1966   | 1802   | 1639   | 1475   | 1311 | 1147 | 983  | 901    | 819  | 737   | 656  | 573   | 469  | 409   | 328  | 246   |  |
| 7x20        | 2.14 x 6.10  | 7864              | 6553   | 5898   | 5242 | 4587   | 4259   | 3932   | 3604   | 3277   | 2949   | 2621 | 2294 | 1966 | 1801   | 1638 | 1473  | 1311 | 1146  | 938  | 818   | 655  | 492   |  |
| 8x10        | 2.44 x 3.05  | 4494              | 3745   | 3370   | 2996 | 2621   | 2434   | 2247   | 2060   | 1873   | 1685   | 1498 | 1311 | 1124 | 1029   | 836  | 842   | 749  | 655   | 562  | 468   | 375  | 281   |  |
| 8x20        | 2.44 x 6.10  | 8987              | 7489   | 6740   | 5991 | 5242   | 4668   | 4493   | 4119   | 3745   | 3370   | 2996 | 2621 | 2247 | 2058   | 1872 | 1684  | 1498 | 1310  | 1123 | 935   | 749  | 562   |  |
| 10x20       | 3.05 x 6.10  | 11234             | 9361   | 8424   | 7489 | 6553   | 6085   | 5617   | 5149   | 4681   | 4213   | 3745 | 3276 | 2808 | 2572   | 2340 | 2105  | 1872 | 1637  | 1404 | 1169  | 936  | 703   |  |

## MEDIDAS Y PESOS TEÓRICOS PTR RECTANGULAR

| Dimensiones<br>pulgadas (mm)                                                             | Color   | Espesor |       |       | Peso  |        | Longitud 6.10 mt |        |
|------------------------------------------------------------------------------------------|---------|---------|-------|-------|-------|--------|------------------|--------|
|                                                                                          |         | cal     | pulg  | mm    | kg/mt | kg/pza | pzas/paq         | kg/paq |
| 1" x 1"<br>(25.4 x 25.4)                                                                 | Azul    | 14      | 0.075 | 1.90  | 1.46  | 8.91   | 96               | 855    |
|                                                                                          | Naranja | 12      | 0.105 | 2.70  | 1.99  | 12.14  | 96               | 1165   |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 2.24  | 13.66  | 96               | 1311   |
|                                                                                          | Verde   | 10      | 0.135 | 3.40  | 2.49  | 15.19  | 96               | 1458   |
| 1 1/4" x 1 1/4"<br>(31.75 x 31.75)                                                       | Azul    | 14      | 0.075 | 1.90  | 1.87  | 11.41  | 80               | 912    |
|                                                                                          | Naranja | 12      | 0.105 | 2.70  | 2.57  | 15.68  | 80               | 1254   |
|                                                                                          | Verde   | 10      | 0.135 | 3.40  | 3.20  | 19.52  | 80               | 1561   |
|                                                                                          |         | 16      | 0.060 | 1.524 | 1.48  | 9.01   | 100              | 901    |
| 1 1/2" x 1 1/2"<br>(38.1 x 38.1)                                                         | Azul    | 14      | 0.075 | 1.90  | 2.27  | 13.85  | 64               | 886    |
|                                                                                          | Naranja | 12      | 0.105 | 2.70  | 3.12  | 19.03  | 64               | 1218   |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 3.50  | 21.35  | 64               | 1366   |
|                                                                                          | Verde   | 10      | 0.135 | 3.40  | 3.91  | 23.85  | 64               | 1526   |
|                                                                                          |         | 16      | 0.060 | 1.524 | 1.84  | 11.22  | 100              | 1122   |
| 3/16" x 3/16"<br>(4.8 x 4.8)<br>2" x 2"<br>(50.8 x 50.8)<br>1/4" x 1/4"<br>(6.35 x 6.35) | Azul    | 14      | 0.075 | 1.90  | 3.02  | 18.42  | 54               | 995    |
|                                                                                          | Naranja | 12      | 0.105 | 2.70  | 4.18  | 25.50  | 54               | 1376   |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 4.75  | 28.98  | 54               | 1564   |
|                                                                                          | Verde   | 10      | 0.135 | 3.40  | 5.31  | 32.39  | 54               | 1748   |
| 2 1/2" x 2 1/2"<br>(64 x 64)<br>1/4" x 1/4"<br>(6.35 x 6.35)                             | Azul    | 14      | 0.075 | 1.90  | 3.80  | 23.18  | 36               | 834    |
|                                                                                          | Naranja | 12      | 0.105 | 2.70  | 5.27  | 32.15  | 36               | 1157   |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 5.80  | 35.38  | 36               | 1273   |
|                                                                                          | Verde   | 10      | 0.135 | 3.40  | 6.17  | 37.64  | 36               | 1354   |
|                                                                                          |         | 9       | 0.148 | 3.80  | 6.75  | 41.18  | 36               | 1481   |
|                                                                                          |         | 8       | 0.165 | 4.20  | 7.44  | 45.38  | 36               | 1633   |
|                                                                                          | Rojo    | 3/16"   | 0.188 | 4.80  | 8.31  | 50.69  | 36               | 1824   |
| 3" x 3"<br>(76 x 76)<br>1/4" x 1/4"<br>(6.35 x 6.35)                                     |         | 14      | 0.075 | 1.90  | 4.35  | 26.54  | 30               | 793    |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 7.06  | 43.07  | 30               | 1291   |
|                                                                                          |         | 10      | 0.135 | 3.40  | 7.54  | 45.99  | 30               | 1379   |
|                                                                                          | Verde   | 9       | 0.148 | 3.80  | 8.25  | 50.33  | 30               | 1509   |
|                                                                                          |         | 8       | 0.165 | 4.20  | 9.10  | 55.51  | 30               | 1664   |
|                                                                                          | Rojo    | 3/16"   | 0.188 | 4.80  | 10.21 | 62.28  | 30               | 1967   |
|                                                                                          |         | 14      | 0.075 | 1.90  | 5.11  | 31.17  | 20               | 623    |
| 3 1/2" x 3 1/2"<br>(89 x 89)                                                             | Blanco  | 11      | 0.120 | 3.00  | 8.32  | 50.75  | 20               | 1014   |
|                                                                                          |         | 10      | 0.135 | 3.40  | 8.89  | 54.23  | 20               | 1084   |
|                                                                                          | Verde   | 9       | 0.148 | 3.80  | 9.73  | 59.35  | 20               | 1186   |
|                                                                                          |         | 8       | 0.165 | 4.20  | 10.76 | 65.64  | 20               | 1312   |
|                                                                                          | Rojo    | 3/16"   | 0.188 | 4.80  | 12.11 | 73.87  | 20               | 1476   |
|                                                                                          |         | 5       | 0.220 | 5.70  | 13.97 | 85.22  | 20               | 1703   |
|                                                                                          | Azul    | 1/4"    | 0.250 | 6.40  | 15.62 | 95.28  | 20               | 1904   |
| 4" x 4"<br>(102 X 102)                                                                   |         | 14      | 0.075 | 1.90  | 5.87  | 35.81  | 20               | 716    |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 9.60  | 58.56  | 20               | 1170   |
|                                                                                          |         | 10      | 0.135 | 3.40  | 10.24 | 62.46  | 20               | 1248   |
|                                                                                          | Verde   | 9       | 0.148 | 3.80  | 11.24 | 68.56  | 20               | 1370   |
|                                                                                          |         | 8       | 0.165 | 4.20  | 12.44 | 75.88  | 20               | 1517   |
|                                                                                          | Rojo    | 3/16"   | 0.188 | 4.80  | 14.00 | 85.40  | 20               | 1707   |
|                                                                                          |         | 5       | 0.220 | 5.70  | 16.19 | 98.76  | 20               | 1974   |
|                                                                                          | Azul    | 1/4"    | 0.250 | 6.40  | 18.15 | 110.72 | 20               | 2213   |
|                                                                                          |         | 14      | 0.075 | 1.90  | 6.63  | 40.44  | 16               | 647    |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 10.24 | 62.46  | 16               | 999    |
|                                                                                          |         | 10      | 0.135 | 3.40  | 11.52 | 70.27  | 16               | 1124   |
|                                                                                          | Verde   | 9       | 0.148 | 3.80  | 13.02 | 79.42  | 16               | 1270   |
|                                                                                          |         | 8       | 0.165 | 4.20  | 14.03 | 85.58  | 16               | 1368   |
|                                                                                          | Rojo    | 3/16"   | 0.188 | 4.80  | 15.90 | 96.99  | 16               | 1551   |
| 4 1/2" x 4 1/2"<br>(114 x 114)                                                           |         | 5       | 0.220 | 5.70  | 18.40 | 112.24 | 16               | 1795   |
|                                                                                          | Azul    | 1/4"    | 0.250 | 6.40  | 20.67 | 126.09 | 16               | 2016   |
|                                                                                          |         | 14      | 0.075 | 1.90  | 7.39  | 45.08  | 9                | 405    |
|                                                                                          | Blanco  | 11      | 0.120 | 3.00  | 12.13 | 73.99  | 9                | 666    |
|                                                                                          |         | 10      | 0.135 | 3.40  | 13.18 | 80.40  | 9                | 723    |
|                                                                                          | Verde   | 9       | 0.148 | 3.80  | 14.22 | 86.74  | 9                | 780    |
|                                                                                          |         | 8       | 0.165 | 4.20  | 16.13 | 98.39  | 9                | 885    |
| 5" x 5"<br>(127 x 127)                                                                   | Rojo    | 3/16"   | 0.188 | 4.80  | 17.79 | 108.52 | 9                | 976    |
|                                                                                          |         | 5       | 0.220 | 5.70  | 20.66 | 126.03 | 9                | 1133   |
|                                                                                          | Azul    | 1/4"    | 0.250 | 6.40  | 23.21 | 141.58 | 9                | 1273   |

# MEDIDAS Y PESOS TEÓRICOS PTR RECTANGULAR

| Dimensiones<br>pulgadas (mm)   | Color   | Espesor |       |      | Peso  |        | Longitud 6.10 mt |        |
|--------------------------------|---------|---------|-------|------|-------|--------|------------------|--------|
|                                |         | cal     | pulg  | mm   | kg/mt | kg/pza | pzas/paq         | kg/paq |
| 2" x 1"<br>(50.8 x 25.4)       | Azul    | 14      | 0.075 | 1.90 | 2.40  | 14.64  | 72               | 1054   |
| 2 1/2" x 1 1/2"<br>(64 x 38.1) | Azul    | 14      | 0.075 | 1.90 | 3.01  | 18.36  | 80               | 1468   |
|                                | Naranja | 12      | 0.105 | 2.70 | 4.16  | 25.38  | 80               | 2029   |
| 3" x 1 1/2"<br>(76.2 x 38.1)   | Azul    | 14      | 0.075 | 1.90 | 3.38  | 20.62  | 40               | 825    |
| 3" x 2"<br>(76 x 50.8)         | Azul    | 14      | 0.075 | 1.90 | 3.80  | 23.18  | 36               | 834    |
|                                | Naranja | 12      | 0.105 | 2.70 | 5.27  | 32.15  | 36               | 1157   |
|                                | Blanco  | 11      | 0.120 | 3.00 | 5.80  | 35.38  | 36               | 1273   |
|                                | Verde   | 10      | 0.135 | 3.40 | 6.17  | 37.64  | 36               | 1354   |
|                                |         | 9       | 0.148 | 3.80 | 6.75  | 41.18  | 36               | 1481   |
|                                |         | 8       | 0.165 | 4.20 | 7.44  | 45.38  | 36               | 1633   |
|                                | Rojo    | 3/16"   | 0.188 | 4.80 | 8.31  | 50.69  | 36               | 1824   |
| 4" x 1 1/2"<br>(101.6 x 38.1)  | Azul    | 14      | 0.075 | 1.90 | 4.17  | 25.44  | 96               | 2441   |
| 4" x 2"<br>(102 x 50.8)        |         | 14      | 0.075 | 1.90 | 4.35  | 26.54  | 30               | 793    |
|                                | Blanco  | 11      | 0.120 | 3.00 | 7.06  | 43.07  | 30               | 1291   |
|                                |         | 10      | 0.135 | 3.40 | 7.54  | 45.99  | 30               | 1379   |
|                                | Verde   | 9       | 0.148 | 3.80 | 8.25  | 50.33  | 30               | 1509   |
|                                |         | 8       | 0.165 | 4.20 | 9.10  | 55.51  | 30               | 1664   |
|                                | Rojo    | 3/16"   | 0.188 | 4.80 | 10.21 | 62.28  | 30               | 1967   |
| 4" x 3"<br>(102 x 76)          |         | 14      | 0.075 | 1.90 | 5.11  | 31.17  | 20               | 623    |
|                                | Blanco  | 11      | 0.120 | 3.00 | 8.32  | 50.75  | 20               | 1014   |
|                                |         | 10      | 0.135 | 3.40 | 8.89  | 54.23  | 20               | 1084   |
|                                | Verde   | 9       | 0.148 | 3.80 | 9.73  | 59.35  | 20               | 1186   |
|                                |         | 8       | 0.165 | 4.20 | 10.76 | 65.64  | 20               | 1312   |
|                                | Rojo    | 3/16"   | 0.188 | 4.80 | 12.11 | 73.87  | 20               | 1476   |
|                                |         | 5       | 0.220 | 5.70 | 13.97 | 85.22  | 20               | 1703   |
|                                | Azul    | 1/4"    | 0.250 | 6.40 | 15.62 | 95.28  | 20               | 1904   |
| 5" x 3"<br>(127 x 76)          |         | 14      | 0.075 | 1.90 | 5.87  | 35.81  | 20               | 716    |
|                                | Blanco  | 11      | 0.120 | 3.00 | 9.60  | 58.56  | 20               | 1170   |
|                                |         | 10      | 0.135 | 3.40 | 10.24 | 62.46  | 20               | 1248   |
|                                | Verde   | 9       | 0.148 | 3.80 | 11.24 | 68.56  | 20               | 1370   |
|                                |         | 8       | 0.165 | 4.20 | 12.24 | 75.88  | 20               | 1517   |
|                                | Rojo    | 3/16"   | 0.188 | 4.80 | 14.00 | 85.40  | 20               | 1707   |
|                                |         | 5       | 0.220 | 5.70 | 16.19 | 98.76  | 20               | 1974   |
|                                | Azul    | 1/4"    | 0.250 | 6.40 | 18.15 | 110.72 | 20               | 2213   |
| 6" x 3"<br>(152 x 76)          |         | 14      | 0.075 | 1.90 | 6.63  | 40.44  | 16               | 647    |
|                                | Blanco  | 11      | 0.120 | 3.00 | 10.24 | 62.46  | 16               | 999    |
|                                |         | 10      | 0.135 | 3.40 | 11.52 | 70.27  | 16               | 1124   |
|                                | Verde   | 9       | 0.148 | 3.80 | 13.02 | 79.42  | 16               | 1270   |
|                                |         | 8       | 0.165 | 4.20 | 14.03 | 85.58  | 16               | 1368   |
|                                | Rojo    | 3/16"   | 0.188 | 4.80 | 15.90 | 96.99  | 16               | 1551   |
|                                |         | 5       | 0.220 | 5.70 | 18.40 | 112.24 | 16               | 1795   |
|                                | Azul    | 1/4"    | 0.250 | 6.40 | 20.67 | 126.09 | 16               | 2016   |
| 6" x 4"<br>(152 x 102)         |         | 14      | 0.075 | 1.90 | 7.39  | 45.08  | 9                | 405    |
|                                | Blanco  | 11      | 0.120 | 3.00 | 12.13 | 73.99  | 9                | 666    |
|                                |         | 10      | 0.135 | 3.40 | 13.18 | 80.40  | 9                | 723    |
|                                | Verde   | 9       | 0.148 | 3.80 | 14.22 | 86.74  | 9                | 780    |
|                                |         | 8       | 0.165 | 4.20 | 16.13 | 98.39  | 9                | 885    |
|                                | Rojo    | 3/16"   | 0.188 | 4.80 | 17.79 | 108.52 | 9                | 976    |
|                                |         | 5       | 0.220 | 5.70 | 20.66 | 126.03 | 9                | 1133   |
|                                | Azul    | 1/4"    | 0.250 | 6.40 | 23.21 | 141.58 | 9                | 1273   |

## NOTAS:

\* TODAS LAS MEDIDAS SE FABRICAN CON ACERO DE CALIDAD COMERCIAL.

\* MEDIDAS MAYORES DE 2" Y CALIBRES MÁS GRUESOS QUE CALIBRE 12 SE OFRECEN EN ESPECIFICACIÓN ASTM A-500 GRADO B.

**SOLERA COMERCIAL**

| Dimensiones |        |            |       | Peso (kg/m) |        |
|-------------|--------|------------|-------|-------------|--------|
| pulgadas    |        | milímetros |       | kg/mt       | kg/pza |
| Espesor     | Ancho  | Espesor    | Ancho |             |        |
| 1/8"        | 1/2"   | 3.2        | 12.7  | 0.32        | 1.92   |
|             | 3/4"   |            | 19.1  | 0.47        | 2.82   |
|             | 1"     |            | 25.4  | 0.63        | 3.78   |
|             | 1 1/4" |            | 31.8  | 0.79        | 4.74   |
|             | 1 1/2" |            | 38.1  | 0.95        | 5.70   |
|             | 2"     |            | 50.8  | 1.27        | 7.62   |
| 3/16"       | 1/2"   | 4.8        | 12.7  | 0.47        | 2.82   |
|             | 3/4"   |            | 19.0  | 0.71        | 4.26   |
|             | 1"     |            | 25.4  | 0.95        | 5.70   |
|             | 1 1/4" |            | 31.8  | 1.19        | 7.14   |
|             | 1 1/2" |            | 38.1  | 1.42        | 8.52   |
|             | 2"     |            | 50.8  | 1.90        | 11.40  |
| 1/4"        | 1/2"   | 6.3        | 12.7  | 0.63        | 3.78   |
|             | 3/4"   |            | 19.0  | 0.95        | 5.70   |
|             | 1"     |            | 25.4  | 1.27        | 7.62   |
|             | 1 1/4" |            | 31.8  | 1.58        | 9.48   |
|             | 1 1/2" |            | 38.1  | 1.90        | 11.40  |
|             | 2"     |            | 50.8  | 2.53        | 15.18  |
|             | 2 1/2" |            | 63.5  | 3.17        | 19.02  |
|             | 3"     |            | 76.2  | 3.80        | 23.18  |
|             | 3 1/2" |            | 88.9  | 4.43        | 26.58  |
| 5/16"       | 4"     | 7.9        | 101.6 | 5.06        | 30.87  |
|             | 2"     |            | 50.8  | 3.17        | 19.02  |
|             | 2 1/2" |            | 63.5  | 3.96        | 23.76  |
|             | 3"     |            | 76.2  | 4.75        | 28.50  |
|             | 3 1/2" |            | 88.9  | 5.54        | 33.24  |
| 3/8"        | 4"     | 9.5        | 101.6 | 6.33        | 37.98  |
|             | 1 1/2" |            | 38.1  | 2.85        | 17.10  |
|             | 2"     |            | 50.8  | 3.80        | 22.80  |
|             | 2 1/2" |            | 63.5  | 4.75        | 28.50  |
|             | 3"     |            | 76.2  | 5.70        | 34.20  |
|             | 3 1/2" |            | 88.9  | 6.67        | 40.64  |
|             | 4"     |            | 101.6 | 7.60        | 45.60  |
| 1/2"        | 6"     | 12.7       | 152.4 | 11.36       | 69.29  |
|             | 1 1/2" |            | 38.1  | 3.80        | 22.80  |
|             | 2"     |            | 50.8  | 5.06        | 30.36  |
|             | 2 1/2" |            | 63.5  | 6.33        | 38.70  |
|             | 3"     |            | 76.2  | 7.60        | 45.60  |
|             | 3 1/2" |            | 88.9  | 8.86        | 53.16  |
|             | 4"     |            | 101.6 | 10.13       | 61.78  |
| 5/8"        | 6"     | 15.9       | 152.4 | 15.18       | 92.59  |
|             | 3"     |            | 76.2  | 9.50        | 57.00  |
|             | 4"     |            | 101.6 | 12.66       | 77.26  |
|             | 6"     |            | 152.4 | 18.97       | 115.72 |
| 3/4"        | 8"     | 19.0       | 203.2 | 25.39       | 154.80 |
|             | 3"     |            | 76.2  | 11.40       | 69.40  |
|             | 3 1/2" |            | 88.9  | 13.33       | 81.32  |
|             | 4"     |            | 101.6 | 15.19       | 92.67  |
|             | 6"     |            | 152.4 | 22.76       | 138.83 |
| 1"          | 8"     | 25.4       | 203.2 | 30.47       | 185.88 |
|             | 3"     |            | 76.2  | 15.19       | 91.14  |
|             | 4"     |            | 101.6 | 20.26       | 121.56 |
|             | 6"     |            | 152.4 | 30.35       | 185.13 |
|             | 8"     |            | 203.2 | 40.63       | 247.84 |

## TUBO CÉDULA 30

| Cédula | Diámetro Nominal |       | Diámetro Ext. | Espesor |       |      | Peso<br>kg/m | Longitud 6.00 mt |        | 6.00 mt<br>kgs/pza | 6.00 mt<br>kgs/pza |
|--------|------------------|-------|---------------|---------|-------|------|--------------|------------------|--------|--------------------|--------------------|
|        | pulg             | mm    |               | Calibre | pulg  | mm   |              | pzas/paq         | kg/paq |                    |                    |
| 30     | 1/2"             | 12.7  | 0.840         | 14      | 0.075 | 1.90 | 0.91         | 127              | 736    | 5.46               | 5.55               |
|        | 3/4"             | 19.0  | 1.050         | 14      | 0.075 | 1.90 | 1.16         | 91               | 675    | 6.96               | 7.08               |
|        | 1"               | 25.0  | 1.315         | 14      | 0.075 | 1.90 | 1.56         | 91               | 857    | 9.36               | 9.52               |
|        | 1 1/4"           | 32.0  | 1.660         | 13      | 0.090 | 2.30 | 2.35         | 61               | 865    | 14.10              | 14.34              |
|        | 1 1/2"           | 38.0  | 1.900         | 13      | 0.090 | 2.30 | 2.71         | 61               | 997    | 16.26              | 16.53              |
|        | 2"               | 51.0  | 2.375         | 12      | 0.105 | 2.70 | 3.96         | 61               | 1457   | 23.76              | 24.16              |
|        | 2 1/2"           | 64.0  | 2.875         | 12      | 0.105 | 2.70 | 4.84         | 37               | 1080   | 29.04              | 29.52              |
|        | 3"               | 76.0  | 3.500         | 11      | 0.120 | 3.00 | 6.75         | 24               | 978    | 40.50              | 41.18              |
|        | 4"               | 102.0 | 4.500         | 11      | 0.120 | 3.00 | 8.74         | 19               | 1002   | 52.44              | 53.31              |

## TUBO CÉDULA 40

| Cédula | Diámetro Nominal |       | Diámetro Ext. | Espesor |       |      | Peso<br>kg/m | Longitud 6.40 mt |        |         |
|--------|------------------|-------|---------------|---------|-------|------|--------------|------------------|--------|---------|
|        | pulg             | mm    |               | Calibre | pulg  | mm   |              | pzas/paq         | kg/paq | kg/pzas |
| 40     | 1/2"             | 12.7  | 0.840         | 12      | 0.105 | 2.70 | 1.27         | 127              | 1041   | 8.13    |
|        | 3/4"             | 19.0  | 1.050         | 12      | 0.105 | 2.70 | 1.68         | 91               | 973    | 10.75   |
|        | 1"               | 25.0  | 1.315         | 10      | 0.135 | 3.40 | 2.65         | 91               | 1544   | 16.96   |
|        | 1 1/4"           | 32.0  | 1.660         | 10      | 0.135 | 3.40 | 3.49         | 61               | 1363   | 22.34   |
|        | 1 1/2"           | 38.0  | 1.900         | 9       | 0.148 | 3.80 | 4.37         | 61               | 1709   | 27.97   |
|        | 2"               | 51.0  | 2.375         | 9       | 0.148 | 3.80 | 5.55         | 61               | 2167   | 32.52   |
|        | 2 1/2"           | 64.0  | 2.875         | 3/16"   | 0.188 | 4.80 | 8.43         | 37               | 1997   | 53.95   |
|        | 3"               | 76.0  | 3.500         | 5       | 0.220 | 5.70 | 12.25        | 24               | 1882   | 78.40   |
|        | 4"               | 102.0 | 4.500         | 5       | 0.220 | 5.70 | 15.98        | 19               | 1944   | 102.27  |
|        | 6"               | 168.0 | 6.625         |         | 0.280 | 7.10 | 29.65        | 7                | 1329   | 189.76  |

TUBERÍA PARA CONDUCCIÓN DE FLUIDOS

| Diámetro Nominal | Diámetro Externo | Espesor |       | CED. No. | kg/m | kgs./pza | Presión de Prueba ASTM-120 |        | Presión de Prueba ASTM-53 |        |
|------------------|------------------|---------|-------|----------|------|----------|----------------------------|--------|---------------------------|--------|
|                  |                  | pulg    | mm    |          |      |          | lbs/pulg2                  | kg/cm2 | lbs/pulg2                 | kg/cm2 |
| 1/2"             | 13               | 0.840   | 21.3  | 0.109    | 2.77 | 40       | 1.27                       | 8.13   | 700                       | 50     |
| 3/4"             | 13               | 0.840   | 21.3  | 0.147    | 3.73 | 80       | 1.62                       | 10.37  | 850                       | 60     |
|                  | 19               | 1.050   | 26.7  | 0.113    | 2.87 | 40       | 1.69                       | 10.82  | 700                       | 50     |
| 1"               | 19               | 1.050   | 26.7  | 0.154    | 3.38 | 80       | 2.20                       | 14.08  | 850                       | 60     |
|                  | 25               | 1.315   | 33.4  | 0.133    | 3.38 | 40       | 2.50                       | 16.00  | 700                       | 50     |
| 1 1/4"           | 25               | 1.315   | 33.4  | 0.179    | 4.55 | 80       | 3.24                       | 20.74  | 850                       | 60     |
|                  | 32               | 1.660   | 42.2  | 0.140    | 3.56 | 40       | 3.39                       | 21.70  | 1000                      | 70     |
| 1 1/2"           | 32               | 1.660   | 42.2  | 0.191    | 4.85 | 80       | 4.47                       | 28.61  | 1500                      | 105    |
|                  | 38               | 1.900   | 48.3  | 0.145    | 3.68 | 40       | 4.05                       | 25.92  | 1000                      | 70     |
| 2"               | 38               | 1.900   | 48.3  | 0.200    | 5.08 | 80       | 5.41                       | 34.62  | 1500                      | 105    |
|                  | 50               | 2.375   | 60.3  | 0.154    | 3.91 | 40       | 5.44                       | 34.82  | 1000                      | 70     |
| 2 1/2"           | 50               | 2.375   | 60.3  | 0.218    | 5.54 | 80       | 7.48                       | 47.87  | 1500                      | 105    |
|                  | 64               | 2.875   | 73.0  | 0.203    | 5.16 | 40       | 8.63                       | 55.23  | 1000                      | 70     |
| 3"               | 64               | 2.875   | 73.0  | 0.160    | 4.06 | X        | 6.90                       | 44.16  | 1000                      | 70     |
|                  | 76               | 3.500   | 88.9  | 0.216    | 5.49 | 40       | 11.29                      | 72.26  | 1000                      | 70     |
| 4"               | 76               | 3.500   | 88.9  | 0.170    | 4.32 | X        | 9.00                       | 57.60  | 1000                      | 70     |
|                  | 102              | 4.500   | 114.3 | 0.237    | 6.02 | 40       | 16.05                      | 102.72 | 1200                      | 85     |
|                  | 102              | 4.500   | 114.3 | 0.188    | 4.78 | X        | 12.91                      | 82.62  | 1000                      | 70     |
|                  |                  |         |       |          |      |          |                            |        | 1500                      | 105    |

## TUBO ESTRUCTURAL LONGITUD ESTÁNDAR 12.20 M

| Diámetro Nominal | Espesor de Pared |       | Cédula | Peso (kg/m) |
|------------------|------------------|-------|--------|-------------|
| 6"               | 3/16"            | 0.188 |        | 19.35       |
|                  | 1/4"             | 0.250 |        | 25.55       |
|                  |                  | 0.280 | 40     | 28.22       |
|                  | 3/8"             | 0.375 |        | 37.20       |
|                  | 1/2"             | 0.500 |        | 48.73       |
| 8"               | 3/16"            | 0.188 |        | 25.37       |
|                  | 1/4"             | 0.250 | 20     | 33.57       |
|                  |                  | 0.277 | 30     | 36.51       |
|                  | 5/16"            | 0.312 |        | 41.14       |
|                  |                  | 0.322 | 40     | 42.65       |
|                  | 3/8"             | 0.375 |        | 49.10       |
|                  |                  | 0.438 |        | 56.94       |
|                  | 1/2"             | 0.500 | 80     | 64.64       |
| 10"              | 3/16"            | 0.188 |        | 31.76       |
|                  | 1/4"             | 0.250 | 20     | 42.09       |
|                  |                  | 0.307 | 30     | 51.03       |
|                  |                  | 0.365 | 40     | 60.50       |
|                  |                  | 0.500 | 60     | 81.55       |
| 12"              | 3/16"            | 0.188 |        | 37.77       |
|                  | 1/4"             | 0.250 | 20     | 50.11       |
|                  |                  | 0.330 | 30     | 65.35       |
|                  |                  | 0.406 | 40     | 79.65       |
|                  | 1/2"             | 0.500 |        | 97.46       |
| 14"              | 3/16"            | 0.188 |        | 41.52       |
|                  | 1/4"             | 0.250 | 10     | 55.11       |
|                  | 5/16"            | 0.312 | 20     | 67.74       |
|                  | 3/8"             | 0.375 | 30     | 81.08       |
|                  |                  | 0.438 | 40     | 94.30       |
|                  | 1/2"             | 0.500 |        | 107.39      |
| 16"              | 3/16"            | 0.188 |        | 47.54       |
|                  | 1/4"             | 0.250 | 10     | 63.13       |
|                  | 5/16"            | 0.312 | 20     | 77.63       |
|                  | 3/8"             | 0.375 | 30     | 92.98       |
|                  | 1/2"             | 0.500 | 40     | 123.30      |
| 18"              | 1/4"             | 0.250 | 10     | 71.12       |
|                  | 3/8"             | 0.375 |        | 104.84      |
|                  | 1/2"             | 0.500 |        | 139.15      |
| 20"              | 1/4"             | 0.250 | 10     | 79.16       |
|                  | 3/8"             | 0.375 | 20     | 116.78      |
|                  | 1/2"             | 0.500 | 30     | 155.12      |
| 24"              | 1/4"             | 0.250 | 10     | 94.53       |
|                  | 3/8"             | 0.375 | 20     | 141.12      |
|                  | 1/2"             | 0.500 |        | 187.06      |

### NOTAS:

\* LA TUBERÍA PARA USO ESTRUCTURAL SE FABRICA BAJO LAS NORMAS  
ASTM A-120, ASTM A-53 Y API 5L GR. B Y X-42.

PERFILES TUBULARES

| Denominación            | Prolamsa<br>Pint./Galv | Regiopytsa<br>Pint./Galv | Arco Metal<br>Pint./Galv | Insa Mex<br>Galvanizado | Peso (largo std. 6.0 m) |             |           | Atados      |             |
|-------------------------|------------------------|--------------------------|--------------------------|-------------------------|-------------------------|-------------|-----------|-------------|-------------|
|                         |                        |                          |                          |                         | 18 (kg/m)               | 18 (kg/pza) | 20 (kg/m) | 20 (kg/pza) | 18 (kg/paq) |
| Tubular 1/2" x 1/2"     | C050                   | 212                      | AC050                    | ZC050                   | 0.47                    | 2.82        | 0.35      | 2.10        | 20          |
| Tubular 3/4" x 3/4"     | C075                   | 210                      | AC075                    | ZC075                   | 0.67                    | 4.02        | 0.53      | 3.18        | 20          |
| Tubular 1" x 1"         | C100                   | 121                      | AC100                    | ZC100                   | 0.93                    | 5.58        | 0.72      | 4.32        | 20          |
| Tubular 1 1/4" x 1 1/4" | C125                   | 208                      | AC125                    | ZC125                   | 1.21                    | 7.26        | 0.91      | 5.46        | 10          |
| Tubular 1 1/2" x 1 1/2" | C150                   | 207                      | AC150                    | ZC150                   | 1.47                    | 8.82        | 1.10      | 6.60        | 10          |
| Tubular 1 3/4" x 1 3/4" | C175                   | 230                      | AC175                    | ZC175                   | 1.71                    | 10.26       | 1.28      | 7.68        | 1           |
| Tubular 2" x 2"         | C200                   | 307                      | AC200                    | ZC200                   | 1.95                    | 11.70       | 1.46      | 8.76        | 8           |
| Tubular 1/2" x 3/8"     | R050                   | *                        | *                        | *                       | 0.38                    | 2.28        |           |             | 20          |
| Tubular 1" x 1/2"       | R100                   | 215                      | AR100                    | ZR100                   | 0.70                    | 4.20        | 0.53      | 3.18        | 20          |
| Tubular 1 1/4" x 3/4"   | R125                   | 228                      | AR125                    | ZR125                   | 0.96                    | 5.76        | 0.72      | 4.32        | 10          |
| Tubular 1 1/2" x 1/2"   | *                      | 128                      | *                        | *                       | 0.94                    | 5.64        |           |             | 100         |
| Tubular 1 1/2" x 1"     | *                      | 229                      | AR229                    | *                       | 1.20                    | 7.20        |           |             | 100         |
| Tubular 1 3/4" x 3/4"   | R175                   | 230                      | AR175                    | ZR175                   | 1.20                    | 7.20        | 0.92      | 5.52        | 10          |
| Tubular 2" x 1"         | R200                   | 216                      | AR200                    | ZR200                   | 1.47                    | 8.82        | 1.11      | 6.66        | 8           |
| Tubular 2 1/4" x 3/4"   | R225                   | 209                      | AR225                    | ZR225                   | 1.47                    | 8.82        | 1.10      | 6.60        | 10          |
| Tubular 2 1/2" x 1 1/4" | R249                   | 241                      | AR249                    | ZR249                   | 1.86                    | 11.16       | 1.39      | 8.34        | 8           |
| Tubular 2 1/2" x 1 1/2" | R250                   | 368                      | AR250                    | ZR250                   | 1.95                    | 11.70       | 1.46      | 8.76        | 1           |
| Tubular 3" x 1 1/4"     | *                      | 223                      | *                        | *                       | 2.00                    | 12.00       | 1.45      | 8.70        | 50          |
| Tubular 3" x 1 1/2"     | R300                   | 322                      | AR300                    | ZR300                   | 2.12                    | 12.72       | 1.60      | 9.60        | 6           |
| Tubular 4" x 1 1/2"     | R400                   | 422                      | AR400                    | ZR400                   | 2.70                    | 16.20       | 2.00      | 12.00       | 4           |

\* EL TUBULAR SE MANEJA EN CALIBRE 18 O CALIBRE 20.

\* (-) PRODUCTOS CON ASTERISCO NO LOS FABRICA DICHO PROVEEDOR.

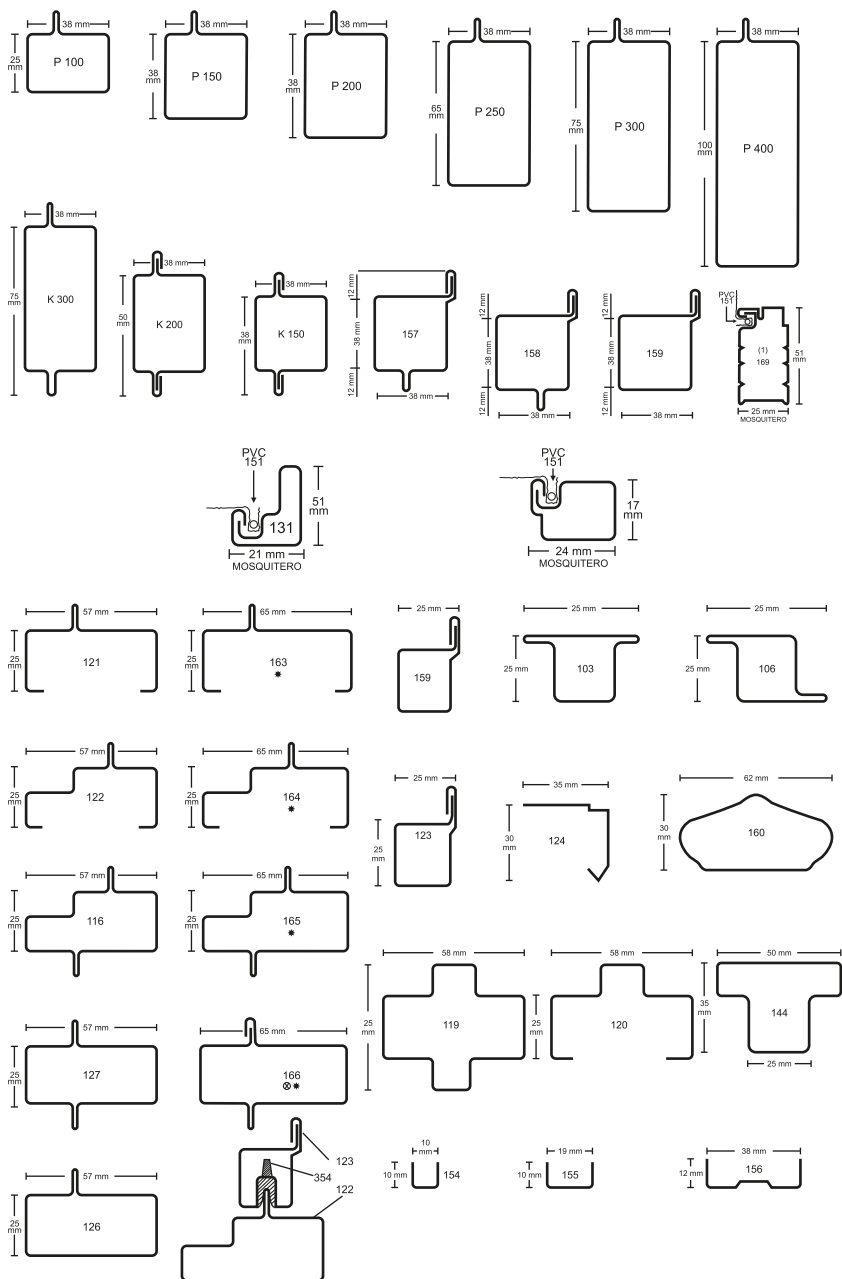
\* EL TABLERO SE FABRICA EN LARGO CONTINUAR O ANCHO CONTINUAR (ACANALADO A LO LARGO O ANCHO).



## PERFILES TUBULARES

| Denominación                   | Prolamsa<br>Pint/Galv | Regiopytsa<br>Pint/Galv | Arco Metal<br>Pint/Galv | Imsa Mex<br>Galvanizado | Peso (largo std. 6.0 m) |             |           |             | Atados   |             |
|--------------------------------|-----------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------|-----------|-------------|----------|-------------|
|                                |                       |                         |                         |                         | 18 (kg/m)               | 18 (kg/pza) | 20 (kg/m) | 20 (kg/pza) | pzas/paq | 18 (kg/paq) |
| Tubular R25                    | P100                  | 236                     | AP100                   | ZP100                   | 1.40                    | 8.40        | 1.01      | 6.06        | 8        | 67.2        |
| Tubular R40                    | P150                  | 234                     | AP150                   | ZP150                   | 1.61                    | 9.66        | 1.23      | 7.38        | 8        | 77.3        |
| Tubular R50                    | P200                  | 261                     | AP200                   | ZP200                   | 1.90                    | 11.40       | 1.41      | 8.46        | 8        | 91.2        |
| Tubular R60                    | P250                  | 268                     | AP250                   | ZP250                   | 2.10                    | 12.60       | 1.58      | 9.48        | 6        | 75.6        |
| Tubular R75                    | P300                  | 310                     | AP300                   | ZP300                   | 2.35                    | 14.10       | 1.73      | 10.38       | 6        | 84.6        |
| Tubular R100                   | P400                  | 410                     | AP400                   | ZP400                   | 2.81                    | 16.86       | 2.11      | 12.66       | 4        | 67.4        |
| Tubular MH25                   | *                     | 237                     | AK100                   | *                       | 1.50                    | 9.00        | 1.16      | 6.96        | 100      | 900.0       |
| Tubular MH40                   | K150                  | 235                     | AK150                   | ZP015                   | 1.90                    | 11.40       | 1.34      | 8.04        | 9        | 102.6       |
| Tubular MH50                   | K200                  | 238                     | AK200                   | ZP020                   | 2.40                    | 14.40       | 1.71      | 10.26       | 10       | 144.0       |
| Tubular MH75                   | K300                  | 239                     | AK300                   | ZP030                   | 2.61                    | 15.66       | 1.91      | 11.46       | 10       | 156.6       |
| Tubular T40 E                  | 159                   | 301                     | A159                    | *                       | 1.80                    | 10.80       | 1.40      | 8.40        | 6        | 64.8        |
| Tubular 121                    | 121                   | 204                     | A121                    | ZV121                   | 1.40                    | 8.40        | 0.95      | 5.70        | 8        | 67.2        |
| Tubular 126                    | 126                   | 243                     | AP126                   | ZV126                   | 1.80                    | 10.80       | 1.31      | 7.86        | 8        | 86.4        |
| Tubular 127                    | 127                   | 242                     | AK127                   | ZV127                   | 1.91                    | 11.46       | 1.51      | 9.06        | 5        | 57.3        |
| Tubular RL                     | 101                   | 201                     | A101                    | ZV101                   | 1.30                    | 7.80        | 0.91      | 5.46        | 10       | 78.0        |
| Tubular RL CORR                | 123                   | 205                     | A123                    | ZV123                   | 1.40                    | 8.40        | 1.04      | 6.24        | 10       | 84.0        |
| Tubular RL MOSQ                | 131                   | 101                     | *                       | ZV131                   |                         |             | 0.67      | 4.02        | 20       | 80.4        |
| Tubular ML                     | 103                   | 219                     | A103                    | ZV103                   | 1.50                    | 9.00        | 1.07      | 6.42        | 10       | 90.0        |
| Tubular ZETA                   | 106                   | 233                     | A106                    | ZV106                   | 1.50                    | 9.00        | 1.01      | 6.06        | 10       | 90.0        |
| Tubular TEE                    | 144                   | 224                     | A144                    | ZV144                   | 1.61                    | 9.66        | 1.21      | 7.26        | 20       | 193.2       |
| Tubular Junquillo 3/8          | 154                   | 114                     | A154                    | ZV154                   | 0.25                    | 1.50        | 0.20      | 1.20        | 50       | 75.0        |
| Tubular Junquillo 3/4          | 155                   | 117                     | *                       | *                       |                         |             | 0.30      | 1.80        | 50       | 90.0        |
| Tubular Pasamanos              | 160                   | 225                     | A160                    | pasamanos               | 1.50                    | 9.00        | 1.12      | 6.72        | 10       | 90.0        |
| Tubular Riel 1400              | 1400                  | 127                     | A1400                   | ZP1400                  |                         |             | 1.22      | 7.32        | 4        | 29.3        |
| Tubular Riel 1500              | 1500                  | 227                     | A1500                   | ZP1500                  | 3.00                    | 18.00       |           | 15.00       | 4        | 72.0        |
| Tubular Zercha                 | *                     | *                       | A30                     |                         | 0.46                    | 2.76        | 2.50      |             | 50       | 138.0       |
| Tubular D60                    | 171                   | 131                     | A171                    | ZD171                   |                         |             | 0.90      | 5.40        | 10       | 54.0        |
| Tubular DD60                   | 170                   | 132                     | A170                    | ZD170                   |                         |             | 1.60      | 9.60        | 10       | 96.0        |
| Tubular Tablero 3x3 pies       | 149                   | 149                     | Tab C CH                | ZT149                   |                         |             | 6.70      | 6.06        | 25       | 1005.0      |
| Tubular Tablero 3x6 pies       | 149                   | 149                     | Tab C CH                | ZT150                   |                         |             | 6.70      | 12.26       | 25       | 1005.0      |
| Tubular Tablero 3x8 pies       | 149                   | 149                     | Tab C CH                | ZT151                   |                         |             | 6.70      | 16.34       | 25       | 1005.0      |
| Tubular Tablero 3x10 pies      | 149                   | 149                     | Tab C CH                | ZT152                   | 1.40                    | 8.40        | 6.70      | 20.43       | 25       | 1005.0      |
| Tubular M225                   | M225                  | 112                     | A129                    | ZM225                   | 2.10                    | 12.60       | 1.10      | 6.60        | 8        | 67.2        |
| Tubular Jamba lados desiguales | M525                  | 145                     | A167                    | ZM225                   | 2.30                    | 13.80       | 1.60      | 9.60        | 7        | 88.2        |
| Tubular Jamba lados iguales    | M600                  | 146                     | A130                    | *                       |                         |             |           |             | 7        | 96.6        |

\* EL TUBULAR SE MANEJA EN CALIBRE 18 O CALIBRE 20.  
\*(-) PRODUCTOS CON ASTERISCO NO LOS FABRICA DICHO PROVEEDOR.  
\* EL TABLERO SE FABRICA EN LARGO CONTINUAR O ANCHO CONTINUAR (ACANALADO A LO LARGO O ANCHO).



## VARILLA CORRUGADA ALTA RESISTENCIA

| No. | Diámetro Nominal |        | Área Nominal    |                   | Peso Unitario |         |        | Perímetro | pzs x tonelada largo std 12 m |
|-----|------------------|--------|-----------------|-------------------|---------------|---------|--------|-----------|-------------------------------|
|     | mm               | pulg   | mm <sup>2</sup> | pulg <sup>2</sup> | kg/m          | kg/pz   | lb/pie |           |                               |
| 2.5 | 7.9              | 5/16"  | 49              | 0.08              | 0.384         | 4.608   | 0.261  | 24.8      | 217                           |
| 3   | 9.5              | 3/8"   | 71              | 0.11              | 0.557         | 6.684   | 0.376  | 29.8      | 150                           |
| 4   | 12.7             | 1/2"   | 127             | 0.20              | 0.996         | 11.952  | 0.668  | 39.9      | 84                            |
| 5   | 15.9             | 5/8"   | 199             | 0.31              | 1.560         | 18.720  | 1.043  | 50.0      | 53                            |
| 6   | 19.1             | 3/4"   | 287             | 0.44              | 2.250         | 27.000  | 1.502  | 60.0      | 37                            |
| 7   | 22.2             | 7/8"   | 387             | 0.60              | 3.034         | 36.408  | 2.044  | 39.7      | 27                            |
| 8   | 25.4             | 1"     | 507             | 0.79              | 3.975         | 47.700  | 2.670  | 79.8      | 21                            |
| 9   | 28.6             | 1 1/8" | 642             | 0.99              | 5.033         | 60.396  | 3.381  | 89.8      | 17                            |
| 10  | 31.8             | 1 1/4" | 794             | 1.23              | 6.225         | 74.700  | 4.172  | 99.9      | 13                            |
| 11  | 34.9             | 1 3/8" | 957             | 1.48              | 7.503         | 90.036  | 5.049  | 109.6     | 11                            |
| 12  | 38.1             | 1 1/2" | 1140            | 1.77              | 8.938         | 107.260 | 6.008  | 119.7     | 9                             |

\* VARILLA CORRUGADA GRADO 42 ASTM A-615

\* LÍMITE DE FLUENCIA MÍNIMA 4200 KG/CM<sup>2</sup>

\* RESISTENCIA A LA TENSIÓN MÍNIMA 5300/CM<sup>2</sup>

## VARILLA CORRUGADA G-6000

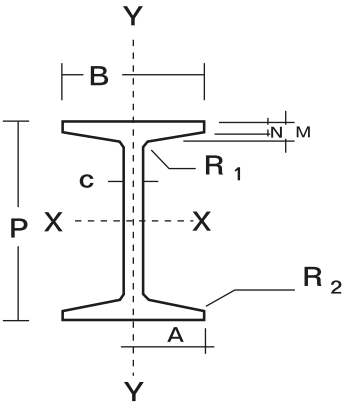
| No. | Diámetro Nominal |       | Área Nominal    |                   | Peso Unitario |       |        | Fuerza Aplicable (kg) | pzs x tonelada largo std 6 mt |
|-----|------------------|-------|-----------------|-------------------|---------------|-------|--------|-----------------------|-------------------------------|
|     | mm               | pulg  | mm <sup>2</sup> | pulg <sup>2</sup> | kg/m          | kg/pz | lb/pie |                       |                               |
|     | 7.94             | 5/16" | 49.5            | 0.08              | 0.388         | 2.33  | 0.26   | 2970                  | 421                           |
|     | 6.35             | 1/4"  | 31.7            | 0.05              | 0.248         | 1.49  | 0.16   | 1900                  | 672                           |
|     | 4.76             | 3/16" | 17.8            | 0.03              | 0.14          | 0.84  | 0.91   | 1069                  | 1151                          |
|     | 3.97             | 5/32" | 12.4            | 0.02              | 0.097         | 0.58  | 0.61   | 742                   | 1661                          |

\* VARILLA CORRUGADA GRADO 6000

\* LÍMITE DE FLUENCIA MÍNIMA 6000 KG/CM<sup>2</sup>

VIGA IPS

| Peralte |      | Peso  | Área  | Eje X-X |      |       | Eje Y-Y |      |      |
|---------|------|-------|-------|---------|------|-------|---------|------|------|
|         |      |       |       | I       | r    | s     | I       | r    | s    |
| mm      | pulg | kg/m  | cm2   | cm4     | cm   | cm3   | cm4     | cm   | cm3  |
| 76.2    | 3    | 8.49  | 10.52 | 103.30  | 3.12 | 27.1  | 19.1    | 1.35 | 6.5  |
| 100.0   | 4L   | 8.34  | 10.60 | 171.00  | 4.01 | 34.2  | 12.2    | 1.07 | 4.8  |
| 101.6   | 4    | 11.46 | 14.26 | 249.30  | 4.17 | 48.9  | 32.1    | 1.50 | 9.5  |
| 127.0   | 5    | 14.88 | 18.52 | 503.30  | 5.21 | 79.3  | 52.1    | 1.65 | 13.4 |
| 152.4   | 6    | 18.60 | 23.29 | 906.80  | 6.25 | 119.0 | 77.0    | 1.83 | 18.2 |
| 177.8   | 7    | 22.78 | 28.90 | 1506.80 | 7.26 | 169.5 | 112.3   | 1.98 | 24.2 |
| 203.2   | 8    | 27.38 | 34.39 | 2367.20 | 8.31 | 233.0 | 157.3   | 2.13 | 31   |

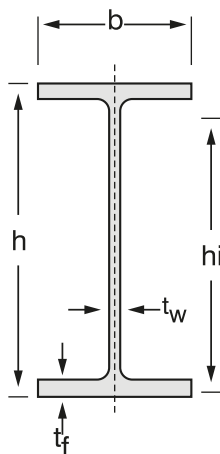


DATOS PARA DETALLAR

| Peralte de la viga | Peso | Peso  |         | Alma    |               |          | Distancia |      |      |      | Gra Mil | Diámetro máximo de los remaches o tornillos |
|--------------------|------|-------|---------|---------|---------------|----------|-----------|------|------|------|---------|---------------------------------------------|
|                    |      | Ancho | Espesor | Espesor | Medio Espesor | Tangente |           |      |      |      |         |                                             |
|                    |      | B     | D       | C       | C/2           | T        | F         | E    | F    | A    |         |                                             |
| mm                 | kg/m | mm    | mm      | mm      | mm            | mm       | mm        | mm   | mm   | mm   | mm      | mm                                          |
| 76.2               | 8.5  | 59.0  | 6.6     | 4.3     | 2.0           | 45.0     | 15.5      | 38.0 | 12.0 | 27.0 | 36.0    | 9.5                                         |
| 101.6              | 11.5 | 68.0  | 7.4     | 4.8     | 3.0           | 70.0     | 16.0      | 50.0 | 13.0 | 32.0 | 38.0    | 12.7                                        |
| 127.0              | 14.9 | 76.0  | 8.3     | 5.3     | 3.0           | 89.0     | 19.0      | 64.0 | 13.0 | 35.0 | 40.0    | 12.7                                        |
| 152.4              | 18.6 | 85.0  | 9.1     | 5.8     | 3.0           | 114.0    | 19.0      | 72.0 | 13.0 | 40.0 | 44.0    | 15.8                                        |
| 203.2              | 27.4 | 102.0 | 10.8    | 6.9     | 4.0           | 159.0    | 22.0      | 76.0 | 14.0 | 48.0 | 56.0    | 19.0                                        |

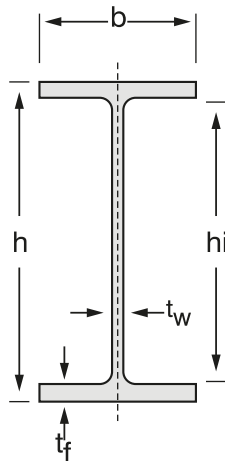
**VIGA IPR**

| Dimensiones<br>pulgadas | Peso   |        | Dimensiones (mm) |     |        |       |       |
|-------------------------|--------|--------|------------------|-----|--------|-------|-------|
|                         | lb/pie | kg/m   | h                | b   | hi     | tw    | tf    |
| 4" X 4"                 | 13.00  | 19.30  | 105.00           | 103 | 88.00  | 7.10  | 8.80  |
|                         | 16.00  | 23.80  | 127.00           | 127 | 109.00 | 6.10  | 9.10  |
|                         | 19.00  | 28.10  | 131.00           | 128 | 109.00 | 6.90  | 10.90 |
| 6" X 4"                 | 8.5    | 12.66  | 148.00           | 100 | 139.00 | 4.30  | 4.90  |
|                         | 9.00   | 13.40  | 150.00           | 100 | 139.00 | 4.30  | 5.50  |
|                         | 12.00  | 17.90  | 153.00           | 102 | 139.00 | 5.80  | 7.10  |
| 6" X 6"                 | 16.00  | 23.80  | 160.00           | 102 | 139.00 | 6.60  | 10.30 |
|                         | 15.00  | 22.30  | 152.00           | 152 | 139.00 | 5.80  | 6.60  |
|                         | 23.00  | 29.80  | 157.00           | 153 | 139.00 | 6.80  | 9.30  |
| 8" X 4"                 | 25.00  | 37.20  | 162.00           | 154 | 139.00 | 8.10  | 11.60 |
|                         | 10.00  | 14.90  | 200.00           | 100 | 190.00 | 4.30  | 5.20  |
|                         | 13.00  | 19.30  | 203.00           | 102 | 190.00 | 5.80  | 6.50  |
| 8" X 5 1/4"             | 15.00  | 22.30  | 206.00           | 102 | 190.00 | 6.20  | 8.00  |
|                         | 14.00  | 20.60  | 203.00           | 133 | 190.00 | 5.00  | 6.40  |
|                         | 18.00  | 26.80  | 207.00           | 133 | 190.00 | 5.80  | 8.40  |
| 8" X 6 1/2"             | 21.00  | 31.30  | 210.00           | 134 | 190.00 | 6.40  | 10.20 |
|                         | 24.00  | 35.70  | 201.00           | 165 | 181.00 | 6.20  | 10.20 |
|                         | 28.00  | 41.70  | 205.00           | 166 | 181.00 | 7.20  | 11.80 |
| 8" X 8"                 | 31.00  | 46.10  | 203.00           | 203 | 181.00 | 7.20  | 11.00 |
|                         | 35.00  | 52.10  | 206.00           | 204 | 181.00 | 7.90  | 12.60 |
|                         | 40.00  | 59.50  | 210.00           | 205 | 182.00 | 9.10  | 14.20 |
| 10" X 4"                | 48.00  | 71.40  | 216.00           | 206 | 181.00 | 10.20 | 17.40 |
|                         | 58.00  | 86.30  | 222.00           | 209 | 181.00 | 13.00 | 20.60 |
|                         | 67.00  | 99.70  | 229.00           | 210 | 182.00 | 14.50 | 23.70 |
| 10" X 5 3/4"            | 12.00  | 17.90  | 251.00           | 101 | 240.00 | 4.80  | 5.30  |
|                         | 15.00  | 22.30  | 254.00           | 102 | 240.00 | 5.80  | 6.90  |
|                         | 17.00  | 25.30  | 257.00           | 102 | 240.00 | 6.10  | 8.40  |
| 10" X 8"                | 19.00  | 28.30  | 260.00           | 102 | 240.00 | 6.40  | 10.00 |
|                         | 16.00  | 23.80  | 253.00           | 145 | 240.00 | 5.00  | 6.40  |
|                         | 22.00  | 32.70  | 258.00           | 146 | 240.00 | 6.10  | 9.10  |
| 12" X 4"                | 26.00  | 38.70  | 262.00           | 147 | 240.00 | 6.60  | 11.20 |
|                         | 30.00  | 44.60  | 266.00           | 148 | 240.00 | 7.60  | 13.00 |
| 12" X 6 1/2"            | 33.00  | 49.10  | 247.00           | 202 | 225.00 | 7.40  | 11.00 |
|                         | 39.00  | 58.00  | 252.00           | 203 | 225.00 | 8.00  | 13.50 |
|                         | 45.00  | 67.00  | 257.00           | 204 | 228.00 | 8.90  | 15.70 |
| 12" X 8"                | 49.00  | 72.90  | 263.00           | 254 | 225.00 | 8.60  | 14.20 |
|                         | 54.00  | 80.40  | 256.00           | 255 | 225.00 | 9.40  | 15.30 |
|                         | 60.00  | 89.30  | 260.00           | 256 | 225.00 | 10.70 | 17.30 |
| 12" X 10"               | 68.00  | 101.20 | 264.00           | 257 | 225.00 | 11.90 | 19.60 |
|                         | 77.00  | 114.60 | 269.00           | 259 | 225.00 | 13.50 | 22.10 |
|                         | 88.00  | 131.00 | 275.00           | 261 | 225.00 | 15.40 | 25.10 |
| 12" X 12"               | 100.00 | 148.80 | 282.00           | 263 | 225.00 | 17.30 | 28.40 |
|                         | 112.00 | 166.70 | 289.00           | 265 | 225.00 | 19.20 | 31.80 |
| 12" X 12 1/2"           | 14.00  | 20.80  | 303.00           | 101 | 292.00 | 5.10  | 5.70  |
|                         | 16.00  | 23.80  | 305.00           | 101 | 292.00 | 5.60  | 6.70  |
|                         | 19.00  | 28.30  | 309.00           | 102 | 291.00 | 6.00  | 6.90  |
| 14" X 5"                | 22.00  | 32.70  | 313.00           | 102 | 291.00 | 6.60  | 10.80 |
|                         | 21.00  | 31.30  | 306.00           | 164 | 291.00 | 5.00  | 7.40  |
|                         | 26.00  | 38.70  | 310.00           | 165 | 291.00 | 5.80  | 9.70  |
| 14" X 6 3/4"            | 30.00  | 44.60  | 313.00           | 166 | 291.00 | 6.60  | 11.20 |
|                         | 35.00  | 52.10  | 317.00           | 167 | 291.00 | 7.60  | 13.20 |
|                         | 40.00  | 59.50  | 303.00           | 203 | 277.00 | 7.50  | 13.10 |
| 14" X 8"                | 45.00  | 67.00  | 306.00           | 204 | 277.00 | 8.50  | 14.60 |
|                         | 50.00  | 74.40  | 310.00           | 205 | 277.00 | 9.40  | 16.30 |
|                         | 53.00  | 78.90  | 306.00           | 254 | 277.00 | 8.90  | 14.80 |
| 14" X 10"               | 58.00  | 86.30  | 310.00           | 254 | 277.00 | 9.10  | 16.30 |
|                         | 65.00  | 96.76  | 308.00           | 305 | 277.00 | 9.90  | 15.40 |
|                         | 72.00  | 107.10 | 311.00           | 306 | 277.00 | 10.90 | 17.00 |
| 14" X 12"               | 79.00  | 117.60 | 314.00           | 307 | 277.00 | 11.90 | 18.70 |
|                         | 87.00  | 129.50 | 318.00           | 308 | 277.00 | 13.10 | 20.60 |
|                         | 96.00  | 142.90 | 323.00           | 309 | 277.00 | 14.00 | 22.90 |
| 14" X 12 1/2"           | 106.00 | 157.70 | 327.00           | 310 | 277.00 | 15.50 | 25.10 |
|                         | 120.00 | 178.60 | 333.00           | 313 | 277.00 | 18.00 | 28.10 |
|                         | 136.00 | 202.37 | 341.00           | 315 | 277.00 | 20.10 | 31.80 |
| 16" X 5"                | 152.00 | 226.20 | 348.00           | 317 | 277.00 | 22.10 | 35.60 |
|                         | 170.00 | 253.00 | 356.00           | 319 | 277.00 | 24.40 | 39.60 |
|                         | 190.00 | 282.80 | 365.00           | 322 | 277.00 | 26.90 | 44.10 |
| 16" X 6 3/4"            | 210.00 | 312.50 | 374.00           | 325 | 277.00 | 30.00 | 48.30 |
|                         | 230.00 | 342.30 | 382.00           | 328 | 277.00 | 32.60 | 52.60 |
|                         | 252.00 | 375.00 | 391.00           | 330 | 277.00 | 35.43 | 57.20 |
| 16" X 8"                | 279.00 | 415.20 | 403.00           | 334 | 277.00 | 38.86 | 62.70 |
|                         | 305.00 | 453.90 | 415.00           | 336 | 277.00 | 41.28 | 68.70 |
|                         | 339.00 | 500.00 | 427.00           | 340 | 277.00 | 45.09 | 75.10 |
| 16" X 10"               | 22.00  | 32.70  | 349.00           | 127 | 332.00 | 5.80  | 6.50  |
|                         | 26.00  | 38.70  | 353.00           | 128 | 332.00 | 6.50  | 10.70 |
|                         | 30.00  | 44.60  | 352.00           | 171 | 332.00 | 6.90  | 9.80  |
| 16" X 12"               | 34.00  | 50.60  | 355.00           | 171 | 332.00 | 7.20  | 11.60 |
|                         | 38.00  | 56.60  | 358.00           | 172 | 332.00 | 7.90  | 13.10 |
|                         | 43.00  | 64.00  | 347.00           | 203 | 320.00 | 7.70  | 13.50 |
| 16" X 14"               | 48.00  | 71.40  | 350.00           | 204 | 320.00 | 8.60  | 15.10 |
|                         | 53.00  | 78.90  | 354.00           | 205 | 320.00 | 9.40  | 16.80 |
|                         | 61.00  | 90.80  | 353.00           | 254 | 320.00 | 9.50  | 16.40 |
| 16" X 16"               | 68.00  | 101.20 | 357.00           | 255 | 320.00 | 10.50 | 18.30 |
|                         | 74.00  | 110.10 | 360.00           | 256 | 320.00 | 11.40 | 19.90 |
|                         | 82.00  | 122.00 | 363.00           | 257 | 320.00 | 13.00 | 21.70 |



## VIGA IPR

| Dimensiones<br>pulgadas | Peso   |        | Dimensiones (mm) |     |        |       |       |
|-------------------------|--------|--------|------------------|-----|--------|-------|-------|
|                         | lb/pie | kg/m   | h                | b   | hi     | tw    | tf    |
| 14" X 14 1/2"           | 90.00  | 133.90 | 356.00           | 369 | 320.00 | 11.20 | 18.00 |
|                         | 99.00  | 147.30 | 360.00           | 370 | 320.00 | 12.30 | 19.80 |
|                         | 109.00 | 162.20 | 364.00           | 371 | 320.00 | 13.30 | 21.80 |
|                         | 120.00 | 178.60 | 368.00           | 373 | 320.00 | 15.00 | 23.90 |
|                         | 132.00 | 196.40 | 372.00           | 374 | 320.00 | 16.40 | 26.20 |
| 14" X 16"               | 145.00 | 215.80 | 375.00           | 384 | 319.00 | 17.27 | 27.70 |
|                         | 159.00 | 236.50 | 380.00           | 395 | 320.00 | 18.92 | 30.20 |
|                         | 176.00 | 261.90 | 387.00           | 398 | 321.00 | 21.09 | 33.30 |
|                         | 193.00 | 287.20 | 393.00           | 399 | 319.00 | 22.61 | 36.60 |
|                         | 211.00 | 314.00 | 399.00           | 401 | 319.00 | 24.89 | 39.60 |
|                         | 233.00 | 346.70 | 407.00           | 404 | 319.00 | 27.18 | 43.70 |
|                         | 257.00 | 382.50 | 416.00           | 406 | 320.00 | 29.85 | 48.00 |
|                         | 283.00 | 421.10 | 425.00           | 409 | 319.00 | 32.77 | 52.60 |
|                         | 311.00 | 462.80 | 435.00           | 412 | 321.00 | 35.81 | 57.40 |
|                         | 342.00 | 509.00 | 448.00           | 416 | 320.00 | 39.12 | 62.70 |
|                         | 370.00 | 550.50 | 455.00           | 418 | 319.00 | 42.04 | 67.60 |
| 16" X 5 1/2"            | 398.00 | 592.30 | 488.00           | 421 | 321.00 | 44.96 | 72.30 |
|                         | 426.00 | 634.00 | 474.00           | 424 | 320.00 | 47.63 | 77.10 |
|                         | 26.00  | 38.70  | 398.00           | 140 | 381.00 | 6.35  | 8.80  |
| 16" X 7"                | 31.00  | 46.10  | 403.00           | 140 | 381.00 | 6.99  | 11.20 |
|                         | 36.00  | 53.60  | 403.00           | 177 | 381.00 | 7.50  | 10.90 |
|                         | 40.00  | 59.50  | 407.00           | 178 | 381.00 | 7.70  | 12.80 |
| 16" X 7"                | 45.00  | 67.00  | 410.00           | 179 | 381.00 | 8.80  | 14.40 |
|                         | 50.00  | 74.40  | 413.00           | 180 | 381.00 | 8.70  | 16.00 |
|                         | 57.00  | 84.80  | 417.00           | 181 | 381.00 | 10.90 | 18.20 |
| 16" X 10 1/4"           | 67.00  | 99.70  | 415.00           | 260 | 381.00 | 10.00 | 16.90 |
|                         | 77.00  | 114.60 | 420.00           | 261 | 381.00 | 11.60 | 19.30 |
|                         | 89.00  | 132.40 | 425.00           | 263 | 381.00 | 13.30 | 22.20 |
|                         | 100.00 | 148.80 | 431.00           | 265 | 381.00 | 14.90 | 25.00 |
| 18" X 6"                | 35.00  | 52.10  | 450.00           | 152 | 428.00 | 7.60  | 10.80 |
|                         | 40.00  | 59.50  | 455.00           | 153 | 428.00 | 8.00  | 13.30 |
|                         | 48.00  | 68.50  | 459.00           | 154 | 428.00 | 9.10  | 15.40 |
| 18" X 7 1/2"            | 41.00  | 61.00  | 450.00           | 189 | 428.00 | 8.10  | 10.80 |
|                         | 45.00  | 67.00  | 454.00           | 190 | 429.00 | 8.50  | 12.70 |
|                         | 50.00  | 74.40  | 457.00           | 190 | 428.00 | 9.00  | 14.50 |
|                         | 55.00  | 81.80  | 460.00           | 191 | 428.00 | 9.90  | 16.00 |
|                         | 60.00  | 89.30  | 463.00           | 192 | 428.00 | 10.50 | 17.70 |
|                         | 65.00  | 96.70  | 466.00           | 193 | 428.00 | 11.40 | 19.00 |
|                         | 71.00  | 105.70 | 469.00           | 194 | 428.00 | 12.60 | 20.60 |
| 18" X 11"               | 76.00  | 113.10 | 463.00           | 280 | 428.00 | 10.80 | 17.30 |
|                         | 86.00  | 128.00 | 467.00           | 282 | 428.00 | 12.20 | 19.60 |
|                         | 97.00  | 144.40 | 472.00           | 283 | 428.00 | 13.60 | 22.10 |
|                         | 106.00 | 157.70 | 476.00           | 284 | 428.00 | 15.00 | 23.90 |
|                         | 119.00 | 177.10 | 482.00           | 285 | 428.00 | 16.70 | 26.90 |
|                         | 130.00 | 193.50 | 489.00           | 283 | 428.00 | 17.00 | 30.50 |
|                         | 143.00 | 212.80 | 495.00           | 285 | 428.00 | 18.50 | 33.50 |
|                         | 158.00 | 235.10 | 501.00           | 287 | 428.00 | 20.60 | 36.60 |
|                         | 175.00 | 260.40 | 509.00           | 289 | 428.00 | 22.60 | 40.40 |
|                         | 192.00 | 285.70 | 517.00           | 291 | 429.00 | 24.38 | 44.50 |
|                         | 211.00 | 314.00 | 525.00           | 293 | 427.00 | 26.92 | 48.50 |
| 21" X 6 1/2"            | 234.00 | 348.20 | 535.00           | 296 | 427.00 | 29.46 | 53.60 |
|                         | 258.00 | 383.90 | 545.00           | 299 | 429.00 | 32.51 | 58.40 |
|                         | 283.00 | 421.10 | 556.00           | 302 | 427.00 | 35.56 | 63.50 |
|                         | 311.00 | 462.80 | 567.00           | 305 | 427.00 | 38.81 | 69.60 |
|                         | 44.00  | 65.50  | 525.00           | 165 | 502.00 | 8.80  | 11.40 |
|                         | 50.00  | 74.40  | 529.00           | 166 | 502.00 | 9.70  | 13.60 |
|                         | 57.00  | 84.80  | 535.00           | 166 | 502.00 | 10.30 | 16.50 |
| 21" X 8 1/4"            | 62.00  | 92.30  | 533.00           | 209 | 502.00 | 10.20 | 15.60 |
|                         | 68.00  | 101.20 | 537.00           | 210 | 502.00 | 10.90 | 17.40 |
|                         | 73.00  | 108.60 | 539.00           | 211 | 501.00 | 11.60 | 18.80 |
|                         | 80.00  | 123.50 | 544.00           | 212 | 502.00 | 13.10 | 21.20 |
|                         | 93.00  | 138.40 | 549.00           | 214 | 502.00 | 14.70 | 23.60 |
| 21" X 12 1/2"           | 101.00 | 150.30 | 543.00           | 312 | 502.00 | 12.70 | 20.30 |
|                         | 111.00 | 165.20 | 546.00           | 313 | 502.00 | 14.00 | 22.20 |
|                         | 122.00 | 181.60 | 551.00           | 315 | 502.00 | 15.20 | 24.40 |
|                         | 132.00 | 196.40 | 554.00           | 316 | 501.00 | 16.50 | 26.30 |
|                         | 147.00 | 218.80 | 560.00           | 318 | 502.00 | 18.30 | 29.20 |
|                         | 166.00 | 247.00 | 571.00           | 315 | 502.00 | 19.00 | 34.50 |
|                         | 182.00 | 270.80 | 577.00           | 318 | 502.00 | 21.10 | 37.60 |
| 24" X 7"                | 201.00 | 299.10 | 585.00           | 319 | 502.00 | 23.10 | 41.40 |
|                         | 55.00  | 81.80  | 598.00           | 178 | 573.00 | 10.00 | 12.80 |
|                         | 62.00  | 92.30  | 603.00           | 179 | 573.00 | 10.90 | 15.00 |
| 24" X 9"                | 68.00  | 101.20 | 603.00           | 228 | 573.00 | 10.50 | 14.90 |
|                         | 76.00  | 113.10 | 608.00           | 228 | 573.00 | 11.20 | 17.30 |
|                         | 84.00  | 125.00 | 612.00           | 229 | 573.00 | 11.90 | 19.60 |
|                         | 94.00  | 139.90 | 617.00           | 230 | 573.00 | 13.10 | 22.20 |
|                         | 103.00 | 153.30 | 623.00           | 229 | 573.00 | 14.00 | 24.90 |
| 24" X 12 3/4"           | 104.00 | 154.80 | 611.00           | 324 | 573.00 | 12.70 | 19.00 |
|                         | 117.00 | 174.10 | 616.00           | 325 | 573.00 | 14.00 | 21.60 |
|                         | 131.00 | 194.90 | 622.00           | 327 | 573.00 | 15.40 | 24.40 |
|                         | 146.00 | 217.30 | 628.00           | 328 | 573.00 | 16.50 | 27.70 |
|                         | 162.00 | 241.10 | 635.00           | 329 | 573.00 | 17.10 | 31.00 |
|                         | 176.00 | 261.80 | 641.00           | 327 | 573.00 | 19.00 | 34.00 |
|                         | 192.00 | 285.70 | 647.00           | 329 | 573.00 | 20.60 | 37.10 |



## VIGA IPR

| Dimensiones<br>pulgadas | Peso   |        | Dimensiones (mm) |     |         |       |       |
|-------------------------|--------|--------|------------------|-----|---------|-------|-------|
|                         | lb/pie | kg/m   | h                | b   | hi      | tw    | tf    |
| 24" X 12 3/4"           | 229.00 | 340.80 | 661.00           | 338 | 573.00  | 24.40 | 43.90 |
|                         | 279.00 | 415.20 | 679.00           | 338 | 573.00  | 29.50 | 53.10 |
|                         | 306.00 | 455.40 | 689.00           | 340 | 573.00  | 32.00 | 57.90 |
|                         | 335.00 | 499.50 | 699.00           | 343 | 573.00  | 35.10 | 63.00 |
|                         | 370.00 | 550.60 | 711.00           | 347 | 573.00  | 38.60 | 69.10 |
| 27" X 10"               | 84.00  | 125.00 | 678.00           | 253 | 645.00  | 11.70 | 16.30 |
|                         | 94.00  | 139.90 | 684.00           | 254 | 646.00  | 12.40 | 18.90 |
|                         | 102.00 | 151.80 | 688.00           | 254 | 646.00  | 13.10 | 21.10 |
|                         | 114.00 | 169.70 | 693.00           | 256 | 646.00  | 14.50 | 23.60 |
|                         | 129.00 | 192.00 | 702.00           | 254 | 646.00  | 15.50 | 27.90 |
| 27" X 14"               | 146.00 | 217.30 | 695.00           | 355 | 645.00  | 15.37 | 24.80 |
|                         | 161.00 | 239.60 | 701.00           | 356 | 647.00  | 16.76 | 27.40 |
|                         | 178.00 | 264.90 | 706.00           | 358 | 646.00  | 18.42 | 30.20 |
|                         | 194.00 | 288.70 | 714.00           | 356 | 646.00  | 19.05 | 34.00 |
|                         | 217.00 | 322.90 | 722.00           | 359 | 646.00  | 21.08 | 38.10 |
|                         | 235.00 | 349.00 | 728.00           | 360 | 646.00  | 23.11 | 40.90 |
|                         | 258.00 | 383.90 | 736.00           | 362 | 646.00  | 24.89 | 45.00 |
|                         | 307.00 | 455.90 | 752.00           | 367 | 646.00  | 29.48 | 53.10 |
|                         | 336.00 | 500.00 | 762.00           | 370 | 646.00  | 32.00 | 57.90 |
| 30" X 10 1/2"           | 368.00 | 547.60 | 772.00           | 372 | 646.00  | 35.05 | 63.00 |
|                         | 99.00  | 147.30 | 753.00           | 265 | 719.00  | 13.20 | 17.00 |
|                         | 108.00 | 160.70 | 758.00           | 266 | 719.00  | 13.80 | 19.30 |
|                         | 116.00 | 172.60 | 762.00           | 267 | 719.00  | 14.40 | 21.60 |
|                         | 124.00 | 184.50 | 766.00           | 267 | 719.00  | 14.90 | 23.60 |
|                         | 132.00 | 196.40 | 770.00           | 268 | 719.00  | 15.60 | 25.40 |
|                         | 148.00 | 220.20 | 779.00           | 269 | 719.00  | 16.50 | 30.00 |
| 30" X 15"               | 173.00 | 257.50 | 773.00           | 381 | 719.00  | 16.64 | 27.10 |
|                         | 191.00 | 284.20 | 779.00           | 382 | 719.00  | 18.03 | 30.10 |
|                         | 211.00 | 314.00 | 786.00           | 384 | 720.00  | 19.69 | 33.40 |
|                         | 235.00 | 349.70 | 795.00           | 382 | 719.00  | 21.08 | 38.10 |
|                         | 261.00 | 388.40 | 803.00           | 385 | 719.00  | 23.62 | 41.90 |
|                         | 292.00 | 434.50 | 813.00           | 387 | 719.00  | 25.91 | 47.00 |
|                         | 326.00 | 485.10 | 823.00           | 390 | 719.00  | 28.96 | 52.10 |
|                         | 357.00 | 531.30 | 833.00           | 393 | 719.00  | 31.50 | 56.90 |
| 33" X 11 1/2"           | 391.00 | 581.90 | 843.00           | 395 | 719.00  | 34.54 | 62.00 |
|                         | 116.00 | 175.60 | 835.00           | 292 | 797.00  | 14.00 | 18.90 |
|                         | 130.00 | 193.50 | 840.00           | 292 | 797.00  | 14.70 | 21.10 |
|                         | 141.00 | 209.80 | 845.00           | 293 | 797.00  | 15.40 | 24.40 |
|                         | 152.00 | 226.20 | 851.00           | 294 | 797.00  | 16.10 | 26.80 |
|                         | 169.00 | 251.50 | 859.00           | 295 | 797.00  | 17.00 | 31.00 |
|                         | 201.00 | 299.10 | 855.00           | 400 | 797.00  | 18.16 | 29.20 |
| 33" X 15 3/4"           | 221.00 | 328.90 | 862.00           | 401 | 798.00  | 19.69 | 32.40 |
|                         | 241.00 | 358.60 | 868.00           | 403 | 796.00  | 21.08 | 35.60 |
|                         | 263.00 | 391.40 | 877.00           | 401 | 797.00  | 22.10 | 39.90 |
|                         | 291.00 | 433.10 | 885.00           | 404 | 797.00  | 24.38 | 43.90 |
|                         | 318.00 | 473.20 | 893.00           | 406 | 797.00  | 26.42 | 48.00 |
|                         | 354.00 | 526.80 | 903.00           | 409 | 797.00  | 29.46 | 53.10 |
|                         | 387.00 | 575.90 | 913.00           | 411 | 797.00  | 32.00 | 57.90 |
|                         | 135.00 | 200.90 | 903.00           | 304 | 863.00  | 15.20 | 20.10 |
| 36" X 12"               | 150.00 | 223.20 | 911.00           | 304 | 863.00  | 15.90 | 23.90 |
|                         | 160.00 | 238.10 | 915.00           | 305 | 863.00  | 16.50 | 25.90 |
|                         | 170.00 | 253.00 | 919.00           | 306 | 863.00  | 17.30 | 27.90 |
|                         | 182.00 | 270.80 | 923.00           | 307 | 863.00  | 18.40 | 30.00 |
|                         | 194.00 | 288.70 | 927.00           | 308 | 863.00  | 19.40 | 32.00 |
|                         | 210.00 | 312.50 | 932.00           | 309 | 863.00  | 21.10 | 34.50 |
|                         | 232.00 | 345.30 | 943.00           | 308 | 863.00  | 22.10 | 39.90 |
|                         | 256.00 | 381.00 | 951.00           | 310 | 863.00  | 24.38 | 43.90 |
| 40" X 12"               | 149.00 | 221.70 | 970.00           | 300 | 928.00  | 16.00 | 21.10 |
|                         | 167.00 | 248.50 | 980.00           | 300 | 928.00  | 16.51 | 26.00 |
|                         | 183.00 | 272.30 | 990.00           | 300 | 928.00  | 16.51 | 30.50 |
|                         | 211.00 | 314.00 | 1000.00          | 300 | 928.00  | 19.05 | 35.90 |
|                         | 235.00 | 349.70 | 1008.00          | 302 | 928.00  | 21.08 | 40.00 |
|                         | 264.00 | 392.90 | 1016.00          | 303 | 928.00  | 24.38 | 43.90 |
|                         | 294.00 | 437.50 | 1026.00          | 305 | 928.00  | 26.92 | 49.00 |
|                         | 327.00 | 486.60 | 1036.00          | 308 | 928.00  | 29.97 | 54.10 |
| 40" X 16"               | 199.00 | 296.10 | 982.00           | 400 | 928.00  | 16.51 | 27.10 |
|                         | 215.00 | 320.00 | 990.00           | 400 | 928.00  | 16.51 | 31.00 |
|                         | 249.00 | 370.60 | 1000.00          | 400 | 928.00  | 19.05 | 36.10 |
|                         | 277.00 | 412.20 | 1008.00          | 402 | 928.00  | 21.08 | 40.00 |
|                         | 297.00 | 442.00 | 1012.00          | 402 | 928.00  | 23.62 | 41.90 |
|                         | 324.00 | 482.20 | 1020.00          | 404 | 928.00  | 25.40 | 46.00 |
|                         | 362.00 | 538.70 | 1030.00          | 407 | 928.00  | 28.45 | 51.10 |
|                         | 372.00 | 553.60 | 1032.00          | 408 | 928.00  | 29.46 | 51.90 |
|                         | 397.00 | 590.80 | 1040.00          | 409 | 928.00  | 30.99 | 55.90 |
| 44" X 16"               | 431.00 | 641.40 | 1048.00          | 412 | 928.00  | 34.04 | 59.90 |
|                         | 230.00 | 342.30 | 1090.00          | 400 | 1088.00 | 18.03 | 31.00 |
|                         | 262.00 | 389.90 | 1100.00          | 400 | 1028.00 | 19.94 | 35.90 |
|                         | 290.00 | 431.60 | 1108.00          | 402 | 1028.00 | 21.97 | 40.00 |
|                         | 335.00 | 498.50 | 1118.00          | 405 | 1028.00 | 26.04 | 45.00 |

# CONVERSIÓN DE UNIDADES

## UNIDADES DE LONGITUD

| Es igual (=)   | mm        | cm       | m        | km        | pulgada  | pie        | yarda     | milla        |
|----------------|-----------|----------|----------|-----------|----------|------------|-----------|--------------|
| 1 milímetro a  | <b>1</b>  | 0.1      | 0.001    | 0.000001  | 0.3937   | 0.00328083 | 0.0010936 | 0.0000062137 |
| 1 centímetro a | 10        | <b>1</b> | 0.01     | 0.00001   | 0.3937   | 0.0328083  | 0.010936  | 0.000062137  |
| 1 metro a      | 1,000     | 100      | <b>1</b> | 0.001     | 39.37    | 3.28083    | 1.0936    | 0.00062137   |
| 1 kilómetro a  | 1,000,000 | 100,000  | 1000     | <b>1</b>  | 39,370   | 3,280.83   | 1,093.60  | 0.62137      |
| 1 pulgada a    | 25.4      | 2.54     | 0.0254   | 0.0000254 | <b>1</b> | 0.08333    | 0.027778  | 0.000016     |
| 1 pie a        | 304.8     | 30.48    | 0.3048   | 0.0003048 | 12       | <b>1</b>   | 0.3333    | 0.0001894    |
| 1 yarda a      | 914.4     | 91.44    | 0.9144   | 0.0009144 | 36       | 3          | <b>1</b>  | 0.0005682    |
| 1 milla a      | 1,609,350 | 160,935  | 1,609.35 | 1.60935   | 63,360   | 5,280      | 1,760     | <b>1</b>     |

## UNIDADES DE PESO

| Es igual (=)         | gr        | kg       | tn         | oz       | lb       | tl         |
|----------------------|-----------|----------|------------|----------|----------|------------|
| 1 gramo a            | <b>1</b>  | 0.001    | 0.000001   | 0.03527  | 0.03937  | 0.00328083 |
| 1 kilogramo a        | 1,000     | <b>1</b> | 0.001      | 35.27    | 0.3937   | 0.0328083  |
| 1 tonelada métrica a | 1,000,000 | 1,000    | <b>1</b>   | 35,270   | 39.37    | 3.28083    |
| 1 onza a             | 28.35     | 0.02835  | 0.00002835 | <b>1</b> | 39.370   | 3,280.83   |
| 1 libra a            | 453.6     | 0.4536   | 0.0004536  | 16       | <b>1</b> | 0.08333    |
| 1 tonelada larga a   | 1,016,050 | 1,016.05 | 1.01605    | 35,836   | 2,240    | <b>1</b>   |

## UNIDADES DE ÁREA

| Es igual (=)            | mm2       | cm2      | m2        | pulg2    | pie2      | yd2       |
|-------------------------|-----------|----------|-----------|----------|-----------|-----------|
| 1 milímetro cuadrado a  | <b>1</b>  | 0.01     | 0.000001  | 0.00155  | 0.0000108 | 0.0000012 |
| 1 centímetro cuadrado a | 1,000     | <b>1</b> | 0.0001    | 0.155    | 0.00108   | 0.00012   |
| 1 metro cuadrado a      | 1,000,000 | 10,000   | <b>1</b>  | 1,550    | 10.76     | 1.196     |
| 1 pulgada cuadrada a    | 645.16    | 6.45160  | 0.0006452 | <b>1</b> | 0.006944  | 0.000772  |
| 1 pie cuadrado a        | 92,903.04 | 929.0304 | 0.0929    | 144      | <b>1</b>  | 0.11111   |
| 1 yarda cuadrada a      | 836,131   | 8,361.31 | 0.8361    | 1,296    | 9         | <b>1</b>  |

## UNIDADES DE VOLUMEN

| Es igual (=)             | cm3       | dm3      | m3         | pulg3    | pie3       | galón      |
|--------------------------|-----------|----------|------------|----------|------------|------------|
| 1 centímetro cúbico a    | <b>1</b>  | 0.001    | 0.000001   | 0.06102  | 0.00003531 | 0.00026417 |
| 1 decímetro cúbico a     | 1,000     | <b>1</b> | 0.001      | 61.02    | 0.3531     | 0.26417    |
| 1 metro cúbico a         | 1,000,000 | 1,000    | <b>1</b>   | 61,020   | 35.31      | 264.17     |
| 1 pulgada cúbica a       | 16.39     | 0.01639  | 0.00001639 | <b>1</b> | 0.0005787  | 0.004329   |
| 1 pie cúbico a           | 28,316.85 | 28.31685 | 0.02831685 | 1,728    | <b>1</b>   | 7.48055    |
| 1 galón EEUU (líquido) a | 3,785.43  | 3.78543  | 0.00378543 | 231      | 0.133663   | <b>1</b>   |

## PESO POR UNIDAD DE LONGITUD

| Es igual (=)            | kg/m     | lb/pie   |
|-------------------------|----------|----------|
| 1 kilogramo por metro a | <b>1</b> | 0.67197  |
| 1 libra por pie a       | 1.48816  | <b>1</b> |

## PESO POR UNIDAD DE SUPERFICIE (ESFUERZOS)

| Es igual (=)              | kg/cm2   | Mpa       | lb/pulg2 |
|---------------------------|----------|-----------|----------|
| 1 kilogramo por cm2 a     | <b>1</b> | 0.0981    | 14.2234  |
| 1 mega pascal a           | 10.1937  | <b>1</b>  | 144.99   |
| 1 libra por pulg2 a [ksi] | 0.07031  | 0.0068971 | <b>1</b> |



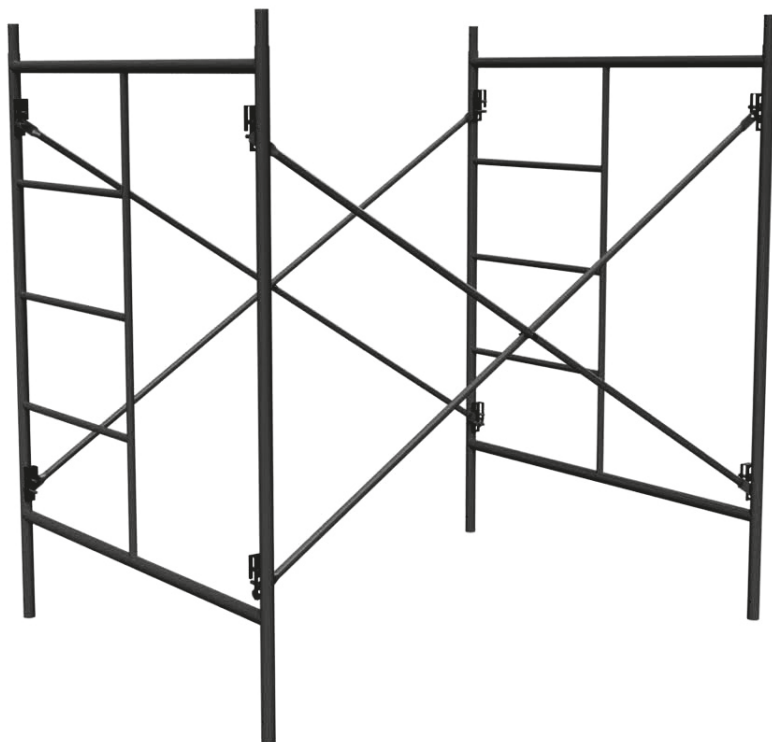
| Fracción | Milésimas | mm     | Cálbre | Milésimas | mm   |
|----------|-----------|--------|--------|-----------|------|
| 1/64"    | 0.016"    | 0.40   | 3      | 0.2391"   | 6.07 |
| 1/32"    | 0.031"    | 0.79   | 4      | 0.2242"   | 5.69 |
| 1/16"    | 0.063"    | 1.59   | 5      | 0.2092"   | 5.31 |
| 1/8"     | 0.125"    | 3.18   | 6      | 0.1943"   | 4.94 |
| 5/32"    | 0.156"    | 3.97   | 7      | 0.1793"   | 4.55 |
| 3/16"    | 0.188"    | 4.76   | 8      | 0.1644"   | 4.18 |
| 1/4"     | 0.250"    | 6.35   | 9      | 0.1495"   | 3.80 |
| 5/16"    | 0.313"    | 7.94   | 10     | 0.1345"   | 3.42 |
| 3/8"     | 0.375"    | 9.53   | 11     | 0.1196"   | 3.04 |
| 7/16"    | 0.438"    | 11.11  | 12     | 0.1046"   | 2.66 |
| 1/2"     | 0.500"    | 12.70  | 13     | 0.0897"   | 2.28 |
| 9/16"    | 0.563"    | 14.29  | 14     | 0.0747"   | 1.90 |
| 5/8"     | 0.625"    | 15.88  | 15     | 0.0673"   | 1.71 |
| 11/16"   | 0.688"    | 17.46  | 16     | 0.0598"   | 1.52 |
| 3/4"     | 0.750"    | 19.05  | 17     | 0.0538"   | 1.37 |
| 13/16"   | 0.813"    | 20.64  | 18     | 0.0478"   | 1.21 |
| 7/8"     | 0.875"    | 22.23  | 19     | 0.0418"   | 1.06 |
| 15/16"   | 0.938"    | 23.81  | 20     | 0.0359"   | 0.91 |
| 1"       | 1.000"    | 25.40  | 21     | 0.0329"   | 0.84 |
| 1 1/8"   | 1.125"    | 28.58  | 22     | 0.0299"   | 0.76 |
| 1 1/4"   | 1.250"    | 31.75  | 23     | 0.0269"   | 0.68 |
| 1 3/8"   | 1.375"    | 34.93  | 24     | 0.0239"   | 0.61 |
| 1 1/2"   | 1.500"    | 38.10  | 25     | 0.0209"   | 0.53 |
| 1 5/8"   | 1.625"    | 41.28  | 26     | 0.0179"   | 0.45 |
| 1 3/4"   | 1.750"    | 44.45  | 27     | 0.0164"   | 0.42 |
| 2"       | 2.000"    | 50.80  | 28     | 0.0149"   | 0.38 |
| 2 1/4"   | 2.250"    | 57.15  | 29     | 0.0135"   | 0.34 |
| 2 1/2"   | 2.500"    | 63.50  | 30     | 0.012"    | 0.30 |
| 2 3/4"   | 2.750"    | 69.85  | 31     | 0.0105"   | 0.27 |
| 3"       | 3.000"    | 76.20  | 32     | 0.0097"   | 0.25 |
| 3 1/2"   | 3.500"    | 88.90  | 33     | 0.009"    | 0.23 |
| 4"       | 4.000"    | 101.60 | 34     | 0.0082"   | 0.21 |

## ANDAMIOS

Marco con 6 pivotes forjados

| Alto | Largo              | Campo                | Tubo  |
|------|--------------------|----------------------|-------|
| 2m   | 1,05mt (o 0.70 mt) | 1,8 mt (o 2,5 - 3mt) | 048mm |

Disponible pintado o galvanizado en caliente.



**ANDAMIOS DE ALTA CALIDAD Y DESEMPEÑO**

## PUNTALES

Ofrecemos una amplia gama de puntales de diferentes versiones y tipologías.

- Puntales de Placa
- Puntales con horquilla simple

| Tubo Interior | Tubo Exterior | Longitud del manguito | Gancho de Seguridad |
|---------------|---------------|-----------------------|---------------------|
| 048mm         | 056mm         | 135mm                 | 014mm               |

- Modelo Disponible 2.20m / 4.00m
- Capacidad de carga de 758 kg
- Pintado de inmersión; libre de óxido
- Mayor durabilidad del mercado
- Factor de seguridad de 1.7 \*

\* Respecto al valor de carga de colapso



### VillaltaSrl

Ponteggi Fav3 Puntelli Tubo & Giunto

Via V. Alfieri, 43 - 35010 Villalta di Gozzio - (PD) - Italia  
Tel. +39 049 9425577 - Fax +39 049 9425707  
www.villaltasrl.it e-mail: info@villaltasrl.it



**Aceros Ocotlán pone a tu disposición una sólida red de sucursales distribuidas estratégicamente en distintas regiones del país. Nuestro objetivo es brindarte un servicio eficiente, con atención profesional, disponibilidad inmediata de materiales y soluciones integrales.**

## **Jalisco**

Acatlán  
Aceros Arandas  
Ajijic  
Altos Arandas  
Altos Matriz  
Atotonilco  
Autlán  
Ayotlán  
Azahares  
Carr. a Morelia  
CAZ Ferretería  
Cedi Ferro  
Cedi Lagos De Moreno  
Central Aceros Ocotlán  
Chapala  
Ciudad Guzmán  
Ciudad Guzmán (Libramiento)  
Cocula  
Degollado  
El Mirador  
El Salto  
Gdl. Matriz  
Jauja  
Jesus María  
La Barca  
La Jalisco  
Lagos De Moreno  
Ocotlán Económicos  
Ocotlán Madero  
Pitillal  
Poncitlán  
Pto. Vallarta Aeropuerto  
Pto. Vallarta Matriz  
Pto. Vallarta Ixtapa  
San Francisco  
San Miguel El Alto  
Tepatitlán 1a  
Tepatitlán 2a  
Tesisitan  
Tonalá  
Zapotlanejo  
Zula Matriz

## **Baja California**

5 Y 10  
Cedi Tijuana  
Ensenada  
Jardín Dorado  
Mexicali

## **Michoacán**

La Piedad  
Lázaro Cárdenas  
Morelia  
Sahuayo

## **Nayarit**

Bucerías  
Ixtlán  
Tepic

## **Nuevo León**

Apodaca  
Láminas Del Norte

## **Yucatán**

Mérida

## **CDMX**

Miguel Hidalgo

## **Coahuila**

Torreón Jumbo  
Torreón Anna  
Torreón Deportiva  
Torreón México  
Torreón Fonseca  
Matamoros

**LO QUE HACE FUERTE A MÉXICO**



# 45 años

## construyendo historias



**45**  
AÑOS  
FORTALECIENDO  
A MÉXICO

 [acerosocotlan.mx](http://acerosocotlan.mx)

**LO QUE HACE**  
**FUERTE**  
**A MÉXICO**